

SOME QUESTIONS IN THE THEORY OF VARIETIES OF GROUPS

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ABSTRACT. In this paper a new method of studying identity relations in groups is described. Among the results obtained by this method is an example of a group variety that does not have an independent basis of identities. A conjecture by P. Hall on marginal subgroups is refuted. The question of the relation between the representation of a group variety in the form of a union and the product of two proper subvarieties is considered. An example of a group variety having a set of continuum many different covering varieties is constructed and a number of results are obtained, as corollaries, on whether (solvable) group varieties can be generated by certain classes of groups.

Bibliography: 26 titles.

At the root of the approach to the questions considered in this paper there lies a modification of the method expounded in [1] and [2]. The essence of this method can be briefly described as follows. To begin with, we consider the relatively free group L of countably infinite rank of a specific center-by-metabelian variety $\mathfrak{L}$.⁽¹⁾ The verbal subgroups $L^4 = x^4(L)$ of L is an elementary abelian 2-group and has the following important property: an additive basis of L^4 (that is, a set of elements generating it independently) has a good description in terms of the set of values of the word x^4 in L (Lemma 3.3). This description is useful of itself, but the situation becomes much more interesting when L is embedded in a larger relatively free group, say in the free group S of the product variety $\mathfrak{L}\mathfrak{R}$ for some variety $\mathfrak{R}$ other than the variety of all groups (for definiteness set $S = F_\infty(\mathfrak{L}\mathfrak{R})$).

If ρ denotes the verbal subgroup of the absolutely free group of countably infinite rank corresponding to $\mathfrak{R}$, then, as is well known, $L \cong \rho(S)$. This gives an additive basis for the abelian verbal subgroup $x^4(\rho(S))$ of S , and by that token is an instrument for the study of the corresponding "layers" of S . The sequence of moves for such a study is usually as follows: the existence of the basis allows us to reduce questions about verbal subgroups of S lying in $x^4(\rho(S))$ to questions about certain subsets of elements of the group $A = S/(\rho(S))^2 = F_\infty(\mathfrak{A}_2\mathfrak{R})$ (see Theorem 4.3); in their turn, these questions reduce, on application of the Magnus-Shmel'kin embedding technique (§6), to questions in the free group $R = S/\rho(S) = F_\infty(\mathfrak{R})$. Thus, for example, the solution of the problem of constructing a group variety not having an independent basis of identities reduces under this scheme to a study of the behavior under endomorphism of certain elements of $R = F_2(\mathfrak{R}_{11})$.

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⁽¹⁾ See §1 for the main definitions and notation.

Here another important property of $\mathfrak{R}$ is used, namely that it has a finite basis for its identities whenever $\mathfrak{R}$ is a finitely based variety.

We come to an account of the contents of the paper and a discussion of the results that we have obtained.⁽²⁾ The distribution of the material among the various sections is as follows: §1 is of an auxiliary nature; in §§2–4 and 6 questions of a basically technical character are considered; §§5, 7 and 8 contain the main results. In §2 we introduce a construction on which the study of the properties of L is based. In §3 the variety $\mathfrak{Q}$ is defined, an additive basis of the subgroup L^4 of $L = F_\infty(\mathfrak{Q})$ is described (Lemma 3.3), and it is shown that the lattice of verbal subgroups of L lying in L^4 is a strictly descending chain (Theorem 3.8). Here, as a first application, we confirm a conjecture of Bryant [3] that a characteristic subgroup of the free group of countably infinite rank of the variety of center-by-metabelian groups is not necessarily fully invariant (see 3.9). It is interesting to compare this fact with a result of Cohen [4], stating that characteristic subgroups of free groups of countably infinite rank of subvarieties of $\mathfrak{A}\mathfrak{R}_C$, in particular of $\mathfrak{A}^2$, are fully invariant. As a corollary, we obtain another example (perhaps about as simple as those in existence; see [3], [5] and [2]) of a characteristic subgroup of the absolutely free group of countably infinite rank that is not fully invariant. An example is the characteristic closure of x^4y^4 .

In §4 we study the free group S (of arbitrary rank) of the product variety $\mathfrak{Q}\mathfrak{R}$ (see 4.1–4.5); in particular, we give a description of an additive basis of the abelian verbal subgroup $x^4(\rho(S))$ of S (see 4.1), and also the lattice of certain verbal subgroups of S (see 4.3 and 4.4) as described. Here we provide simple examples of relatively free groups defined by a finite number of identities but not having finite bases for their identities.

§5 begins with a module interpretation of the verbal subgroups of S lying in $x^4(\rho(S))$. A consideration of properties of this interpretation concludes a description of the “body” of the method, and the remainder of the section is devoted to applications. The scheme of the application is conveniently investigated by the construction of an example of a 2-generator relatively free group that is solvable of length 5 but not residually finite (see 5.2). Further we give a refutation of a conjecture of P. Hall. Generalizing the well-known Schur-Baer theorem, Hall [6] made the following conjecture: for every word v and every group G , if the index $|G: v^*(G)| = m$ of the corresponding marginal subgroup is finite, then the order of the verbal subgroup $v(G)$ is finite and divides some power of m . Hall himself verified this in the class of almost polycyclic groups. From the arguments used by Hall [6], it is easy to deduce that the conjecture is true for any variety of groups such that every subvariety is generated by finite groups. Merzlyakov verified the conjecture for linear groups in [7]. It is proved in Theorem 5.3 that for every prime p there exists a word $v_p(x, y)$ and a relatively free group G that is solvable of length 5 and such that $|G: v_p^*(G)| = p^2$ but $|v_p(G)| = 2$.

In §6 commutator calculus in free nilpotent groups and the Magnus-Shmel'kin embedding technique are adapted to a discovery of the following interesting effect (Theorem 6.10): for odd $c \geq 9$, the verbal subgroup $x^4(\gamma_{c+1}(S))$ of $S = F_\infty(\mathfrak{Q}\mathfrak{R}_c)$ contains an infinite direct product of verbal subgroups of S . In contrast we recall that the lattice of verbal subgroups of $L = \gamma_{c+1}(S)$ lying in L^4 is a descending chain. From 6.10 we also get the following curious (but subsequently useful) Corollary 6.11: there exists a 2-generator

⁽²⁾ The main results were announced in [26].

relatively free group $S_1 \in \mathfrak{A}^7$ having an infinite verbal subgroup W that is an elementary abelian 2-group and such that every subgroup of W is verbal in S_1 .

Assertions 6.10 and 6.11 lie at the root of the results obtained in §7. The first of these results was stimulated by the not completely successful formulation of the first part of Problem 7 of Hanna Neumann's book [8], concerning the connection between the representation of a variety of groups as a product and as a union of two subvarieties (for a detailed discussion of this question see [9]). We say that a decomposition of a variety as a union $\mathfrak{L} = \mathfrak{M} \cup \mathfrak{N}$ is *minimal* if strict inclusions $\mathfrak{M}' \subset \mathfrak{M}$ and $\mathfrak{N}' \subset \mathfrak{N}$ imply strict inclusions $\mathfrak{M}' \cup \mathfrak{N}' \subset \mathfrak{L} \subsetneq \mathfrak{M} \cup \mathfrak{N}'$. Using Zorn's lemma, it is easy to show that every decomposition can be extended to a minimal one. The number of different minimal decompositions of a variety characterizes the number of essentially different decompositions of it as a union of two subvarieties. The following two questions were posed in [9]. Suppose that the group variety $\mathfrak{G}$ cannot be decomposed as a union of two proper subvarieties, and let $\mathfrak{R}$ be any variety of groups; can $\mathfrak{G}\mathfrak{R}$ be decomposed as a union? Let $\mathfrak{G}$ be any nontrivial variety; does every minimal proper decomposition $\mathfrak{M} \cup \mathfrak{N}$ of $\mathfrak{G}\mathfrak{R}$ follow from a proper decomposition of $\mathfrak{G} = \mathfrak{M}' \cup \mathfrak{N}'$ (in the sense that $\mathfrak{M} = \mathfrak{M}'\mathfrak{R}$ and $\mathfrak{N} = \mathfrak{N}'\mathfrak{R}$)? The positive answer to the first question was obtained by Kovács [10] in the case where $\mathfrak{G}$ is abelian or $F_\infty(\mathfrak{G})$ does not have nontrivial verbal abelian subgroups. In general the answer to the first question (and thus to the second) is negative. Indeed, there exists a center-by-metabelian variety $\mathfrak{L}'$ of exponent 8 that is not decomposable as a union of any two proper subvarieties, and such that $\mathfrak{L}'\mathfrak{R}_9$ has continuum many different minimal decompositions as a union of two proper subvarieties (Theorem 7.1).

In Theorem 7.2 we provide an example of a countable system of 2-variable identities having the following property: every subsystem of not more than countable cardinal τ fails to be equivalent to any system of 2-variable identities of cardinal $< \tau$.

The following circle of questions concerns covering elements in the lattice of varieties of groups and the generation of varieties by certain classes of groups. As Ol'shanskii [11] observed, every variety of groups has a covering variety. Using his arguments, it is not hard to show that every variety of groups has an infinite set of covering varieties. It is also easy to see that all Burnside varieties $\mathfrak{B}_n$ ($n \geq 1$), and also subvarieties of locally finite Burnside varieties ($n = 3, 4, 6$) have exactly a countable set of covering varieties. Besides this, it seems that little is known for groups. On the other hand, in the variety of semigroups there are examples of varieties having continuum many different covering varieties (see [12]). In Theorem 7.3, we provide an example of a subvariety of $\mathfrak{A}^7$ having a continuum of covering varieties in $\mathfrak{A}^7$.

The problem of the cardinality of the set of varieties covering a given variety is of interest in itself, but also has useful applications. These applications use the fact that, if $\{\mathfrak{M}_\alpha\}$ is family of continuum many different varieties covering a variety $\mathfrak{M}$, then $\{\mathfrak{M}_\alpha \setminus \mathfrak{M}\}$ is a family of continuum many nonempty pairwise disjoint classes of groups. Thus 7.3 gives the important Corollary 7.4: for any class $\mathfrak{K}$ of groups of cardinality less than that of the continuum, there exist continuum many varieties of solvable groups of length 7 with one and the same stock of $\mathfrak{K}$ -groups (groups lying in $\mathfrak{K}$) and thus not generated by $\mathfrak{K}$ -groups. In particular, this gives a negative answer to Ol'shanskii's question (see [13], 4.47) on the existence of a countable system of groups such that every variety of groups is generated by some subsystem. (As is clear from 7.4, such a system fails to exist even in $\mathfrak{A}^7$.) As a second corollary we obtain the existence for the class $\mathfrak{K}$ mentioned

above of a 2-generator relatively free group that is solvable of length 7 and not residually in $\mathcal{K}$. §7 concludes with an answer to a problem of Kuznetsov (see [13], 6.17): there exist varieties of solvable groups of length 7 not generated by their recursively presented groups, and consequently not by groups with solvable word problem. This last fact is a good comment on the fact that varieties in which Ol'shanskiĭ's conjecture is false are "wild" from the point of view of complication of the structure of their subvarieties. It is curious also to note the following. It is clear that every recursively based variety (that, a variety having a recursively enumerable basis for its laws) is generated by recursively presented groups. On the other hand, Theorem 1 of [1] gives an example of a 2-generator group with solvable word problem generating a variety of groups that is not recursively based. In concluding our discussion of the results of §7, we note that, using more "primitive" methods, 7.3–7.6 can be verified already in $\mathfrak{A}^4$.

Shortly after Ol'shanskiĭ [14] proved that not every variety of groups is finitely based, and Adyan [15] and Vaughan-Lee [16] constructed the first examples of infinite independent systems of group laws, Tarski posed the following problem (see [13], 4.64): does every variety of groups have an independent basis for its identities? Ol'shanskiĭ [11] proved the only result in this direction: every subvariety of a finitely based locally finite variety of groups is independently based. In particular, every solvable locally finite variety has an independent basis. In Theorem 8.5, we give an example of a variety of solvable groups of length 7 not having an independent basis. Thus the answer to Tarski's problem is in the negative. It also proves the necessity of the condition on local finiteness in Ol'shanskiĭ's result. Earlier negative answers to the analogous result in the varieties of semigroups and inverse semigroups were obtained by Trakhtman [17] and Kleĭman [18]. The variety of Theorem 8.5 can be given explicitly by identities in less than 100 variables. It is curious to note that this variety lies in $\mathfrak{N}_2 \mathfrak{A} \mathfrak{N}_{11}$, whereas every subvariety of $\mathfrak{A} \mathfrak{N}_C$ is independently based. This latter result follows from [11], insofar as the proof of the result of Ol'shanskiĭ quoted above uses only the fact that every subvariety of a locally finite variety is generated by finite groups; that is, the condition on local finiteness can be replaced by the assumption that every subvariety is generated by finite groups. We add that the variety figuring in Theorem 8.5 is the intersection of two independently based varieties.

In passing in the proof of Theorem 8.5, we obtain a refinement of a result in [2]. In answer to a problem of Ol'shanskiĭ (see [13], 5.42), it was proved in assertion 7.5 of [2] that, for sufficiently large n , there is an infinite sequence $\{v_i\}$ of words in the free group F_n of rank n whose fully invariant closures form a strictly increasing chain; that is, $1 \subsetneq v_1(F_n) \subsetneq v_2(F_n) \subsetneq \dots$. It is proved in Corollary 8.11 that we can choose $n = 89$. Certain additional arguments lower the value of n to 73. It is unknown whether this phenomena happens in F_2 .

The second result in §8, Theorem 8.2, gives an example of two varieties $\mathfrak{M} \subset \mathfrak{M}_1 \subset \mathfrak{A}^7$ such that $\mathfrak{M}$ has an infinite independent basis of identities, but does not have a covering variety in $\mathfrak{M}_1$, and thus is not defined in $\mathfrak{M}_1$ by an independent system of identities (cf. Lemma 8.1). It is interesting to note that the proof of Theorem 8.2 gives incidentally a kind of verbal analogue of the infinite quasicyclic group, in that in $G = F_2(\mathfrak{M}_1)$ there exists an infinite verbal subgroup W such that every verbal subgroup of G lying in W and different from W is finite and is the fully invariant closure of a single element of G , while the set of all verbal subgroups of G lying in W forms an (infinite) ascending chain under inclusion.

In conclusion, the author expresses his deep gratitude to his supervisors and to the participants in the research seminar on group theory at Moscow State University, at whose meetings the results of this paper were discussed.

§1. Definitions, notation and auxiliary results. ⁽³⁾

1°. By F_∞ , F and F_n we understand the absolutely free groups of countably infinite, arbitrary and finite rank n respectively; $F_\infty(\mathfrak{R})$, $F(\mathfrak{R})$ and $F_n(\mathfrak{R})$ denote the corresponding things for a variety $\mathfrak{R}$. Let $\{x_1, x_2, \dots\}$ and $\{y_1, y_2, \dots\}$ to two arbitrary countable infinite alphabets, and let $v(x) = v(x_1, \dots, x_n)$ be a word in the alphabet of x 's. Then by $v(y)$ we understand a word in the y -alphabet obtained from $v(x)$ by the transformation $x_i \rightarrow y_i$ ($i = 1, 2, \dots$); that is, $v(y) = v(y_1, \dots, y_n)$. In the sequel we fix an alphabet $t_1, t_2, \dots$ as a system of free generators of F_∞ and set $F_n = \text{gp}(t_1, \dots, t_n)$ (by $\text{gp}(M)$ we mean the subgroup generated by the set M). It will usually be clear from the context in which alphabet one or another word is written. Thus, for example, an arbitrary element $v \in F_n$ will be denoted by $v = v(t) = v(t_1, \dots, t_n)$. The letters t_i can be considered as variables; then a *value* of v in an arbitrary group G is an element $g \in G$ of the form $v(g_1, \dots, g_n)$, $g_i \in G$. The set of all values of v in G is denoted by $v[G]$, and the corresponding *verbal* subgroup is $v(G)$. By definition $v(G) = \text{gp}(v[G])$. The *identity* $v(t) \equiv 1$ holds in G if $v[G] = 1$ (sometimes in this case we use other variables).

Philip Hall introduced the concept of the *marginal* subgroup $v^*(G)$ of G :

$$\begin{aligned} v^*(G) &= \{ g \in G \mid v(g_1, \dots, g_n) = v(g_1 g, g_2, \dots, g_n) = v(g_1, g_2 g, g_3, \dots, g_n) \\ &= \dots = v(g_1, \dots, g_{n-1}, g_n g); \forall g_1, \dots, g_n \in G \}. \end{aligned}$$

A marginal subgroup is characteristic, and a basic property is that the values of v in G depend only on the cosets of $G \bmod v^*(G)$. For example, if $v = [t_1, t_2]$ then the verbal subgroup $v(G) = G'$ corresponds to the marginal subgroup $v^*(G)$, the center of G in this case.

Let G be a relatively free group freely generated by a set $\{x_1, x_2, \dots\}$. Since G is free in the variety $\text{var } G$ it generates, the letters x_i can play the role of variables, which is well illustrated by the fact that $v[G]$ coincides in this case with the set of all images of $v(x)$ under endomorphisms of G ; that is, $v[G] = \{v(x)\varphi \mid \varphi \in \text{End } G\}$. This fact we shall use constantly without explicit statement. If H is a relatively free group freely generated by $\{y_1, y_2, \dots\}$, then for every homomorphism $\varphi \in \text{Hom}(G, H)$ we can write

$$x_i \varphi = y_1^{\varphi_{i1}} \dots y_n^{\varphi_{in_i}} \cdot h_i, \quad \text{where } h_i \in H' \text{ and } \varphi_{ij} \in \mathbb{Z}.$$

We denote the corresponding matrix by $(\varphi) = (\varphi_{ij})$. If the ranks of G and H are finite and equal, we write the determinant $\det((\varphi_{ij}))$ as $|\varphi|$.

For an arbitrary natural number n , we denote by $\exp_2 n$ the exponent of the power of 2 in the decomposition of n into prime factors. By definition, $\exp_2 0 = \infty$.

We recall some standard notation. $\mathfrak{A}$ is the variety of all abelian groups, $\mathfrak{A}^n$ the variety of all solvable groups of length $\leq n$, $\mathfrak{B}_n$ the Burnside variety of exponent n (that is, the

⁽³⁾ For notation and concepts relating to varieties of groups and not mentioned here, see [8].

variety defined by the law $x^n \equiv 1$), $\mathfrak{A}_n = \mathfrak{B}_n \cap \mathfrak{A}$, $\mathfrak{N}_c$ the variety of all nilpotent groups of class $\leq c$, $\gamma_c(G)$ the c th term of the lower central series of G , $G^n = x^n(G)$, $\mathbb{Z}$ the ring of integers and $\mathbb{Z}_2$ the field of two elements.

Let R be a ring with identity, S a semigroup with identity and S_1 a subsemigroup of S containing the identity. We denote the semigroup ring of S over R by $R(S)$, and the two-sided ideal of $R(S)$ generated by the set $\{s - 1 | s \in S_1\}$ by Δ_{S_1} . In particular, Δ_S denotes the augmentation ideal of $R(S)$.

A group is said to be *center-by-metabelian* if its central factor-group is metabelian. This is equivalent to the assumption that the identity $[[x, y], [z, t], u] \equiv 1$ holds.

A system of identities is said to be *independent* (in a variety $\mathfrak{R}$) if it is not equivalent (in $\mathfrak{R}$) to any proper subsystem. The variety $\mathfrak{M}$ is *independently based* (in $\mathfrak{R}$) if it has (in $\mathfrak{R}$) an independent basis for its identities. A system of identities in n variables of finite or countably infinite cardinality τ is said to be *n-independent* (in $\mathfrak{R}$) if no subsystem of cardinality $\tau_0 \leq \tau$ is equivalent (in $\mathfrak{R}$) to a system of identities in variables of cardinality $< \tau_0$. In Theorem 7.2 we give an example of an infinite system of identities that is 2-independent in $\mathfrak{A}^7$.

We say that a relatively free group G is defined by a finite number of identities if $G = F/V$, where V is the verbal subgroup of F defined by a finite set of words. In other words, G is a free group of some finitely based variety. For a group of infinite rank this is equivalent to the fact that $\text{var}(G)$ is finitely based, while for a group of finite rank this equivalence does not hold (see 4.8).

If a variety $\mathfrak{M}$ is a proper maximal subvariety of a variety $\mathfrak{R}$, then $\mathfrak{R}$ is said to be a *covering* for $\mathfrak{M}$. See Lemma 8.1 for the connection between this concept and that of being independently based.

2°. We state now some known results.

LEMMA 1.1 ([8], 31.62 and 32.21). *Free nilpotent groups are torsion-free. Every finitely generated torsion-free nilpotent group is residually a finite p -group for every prime p .*

LEMMA 1.2 (Baumslag [8], 42.35). *Suppose that free groups of a variety $\mathfrak{M}$ are residually finite p -groups for an infinite set of primes p . Then every subset of $F(\mathfrak{M})$ freely generating modulo $F(\mathfrak{M})'$ a free subgroup of $F(\mathfrak{M})/F(\mathfrak{M})'$ generates a subgroup of $F(\mathfrak{M})$ $\mathfrak{M}$ -freely.*

LEMMA 1.3 (Baumslag [8], 42.43). *Suppose that free groups of varieties $\mathfrak{M}$ and $\mathfrak{R}$ are residually finite p -groups for one and the same prime p , and let ρ be the verbal subgroup of F_∞ corresponding to $\mathfrak{R}$. Then every subset of $K = F(\mathfrak{M}\mathfrak{R})$ freely generating modulo $\rho(K)$ an $\mathfrak{R}$ -free group freely generates an $\mathfrak{M}\mathfrak{R}$ -free subgroup of K .*

§2. Construction of the group $C(G, N, M)$

The material in this section (apart from Lemma 2.2) is taken from [2], and is included here because of the importance of the construction introduced there, and also for completeness of exposition.

Let G be a group, $G \triangleright N$, and N an elementary abelian 2-group. We choose some nonempty set of nonidentity elements $M \subset N$ such that $M^g = M$ ($\forall g \in G$). We denote by $\bar{g}$ the image of g in G/N , and construct the group $B = B(G, N, M)$ generated by all

symbols of the form b^g and b_*^z for $g \in G$ and $z \in G/N$ with defining relations of the form

$$\begin{aligned} (b^g)^2 &= (b_*^z)^2 = 1, \\ [b^g, b^h] &= 1 \quad \text{for } gh^{-1} \notin M, \\ [b^g, b^h] &= b_*^{\bar{g}} \quad \text{for } gh^{-1} \in M, \\ [b^g, b_*^z] &= 1 \quad \text{for arbitrary } g, h \in G \text{ and } z \in G/N. \end{aligned}$$

It is easy to see that $B = \text{gp}(b^g | g \in G)$ and that $B' = \text{gp}(b_*^z | z \in G/N)$ is an elementary abelian 2-group in the center of B with independent generators $\langle b_*^z | z \in G/N \rangle$. It follows in particular from this that $B \in \mathfrak{A}_2^2 \cap \mathfrak{N}_2$. A construction of this type first appeared in a paper of Vaughan-Lee [16]. Since B' is isomorphic to the additive group of the group ring $\mathbf{Z}_2(G/N)$, it is correct to write $b_*^{z_1} \cdots b_*^{z_n} \stackrel{\text{def}}{=} b_*^{z_1 + \cdots + z_n}$ ($z_i \in G/N$). Thus we can assume that every element of B' has the form b_*^α , with $\alpha \in \mathbf{Z}_2(G/N)$. In this notation the operation in B' appears as $b_*^{\alpha_1} b_*^{\alpha_2} = b_*^{\alpha_1 + \alpha_2}$, $\alpha_i \in \mathbf{Z}_2(G/N)$.

There is a natural action of G on B :

$$(b^g)^h = b^{gh}, \quad (b_*^\alpha)^h = b_*^{\alpha h}, \quad \text{where } g, h \in G \text{ and } \alpha \in \mathbf{Z}_2(G/N).$$

For $C = C(G, N, M)$ we take the splitting extension of B by G with the given action. We define now the characteristic function χ of the subset M of G , setting for each $g \in G$

$$\chi(g) = \begin{cases} 1, & \text{if } g \in M, \\ 0, & \text{if } g \notin M. \end{cases}$$

We note that the choice of M gives that

$$\chi(h^{-1}gh) = \chi(g) \quad (g, h \in G).$$

We choose the element $f = ab^{g_1} \cdots b^{g_n}$ in C , where $a \in N$ and $g_i \in G$.

LEMMA 2.1. $f^4 = (b_*^{\bar{g}_1 + \cdots + \bar{g}_n})^{\chi(a)}$, so that $f^4 = 1 \Leftrightarrow$ either $a \notin M$ or the number of elements g_i occurring in the expression for f such that $\bar{g}_i = z$ is even for arbitrary $z \in G/N$.

PROOF. We square f twice:

$$f^2 = b^{g_1 a} \cdots b^{g_n a} b^{g_1} \cdots b^{g_n}.$$

Since B' is elementary abelian and central in B , before squaring the factors can be rearranged as follows:

$$f^4 = (b^{g_1 a} b^{g_1} \cdots b^{g_n a} b^{g_n})^2.$$

It is easy to see that the pairs $b^{g_i a} b^{g_i}$ and $b^{g_j a} b^{g_j}$ commute. In fact, if $\bar{g}_i \neq \bar{g}_j$ this follows immediately from the relations of B . If $\bar{g}_i = \bar{g}_j$, then because expansion of commutators is distributive, we get

$$\begin{aligned} [b^{g_i a} b^{g_i}, b^{g_j a} b^{g_j}] &= [b^{g_i a}, b^{g_j a}][b^{g_i a}, b^{g_j}] [b^{g_i}, b^{g_j a}] [b^{g_i}, b^{g_j}] \\ &= (b_*^{\bar{g}_i})^{\chi(g_i g_j^{-1}) + \chi(g_i a g_j^{-1}) + \chi(g_i a g_j^{-1}) + \chi(g_i g_j^{-1})} = 1. \end{aligned}$$

The desired relation now follows from the identity

$$(b^{g_i a} b^{g_i})^2 = [b^{g_i}, b^{g_i a}] = (b_*^{\bar{g}_i})^{\chi(g_i a g_i^{-1})} = (b_*^{\bar{g}_i})^{\chi(a)}.$$

This proves the lemma.

We consider now the subgroup $E = N \cdot B$ of C .

LEMMA 2.2. $E \in \mathfrak{A}_3^3$, and also E satisfies the identity $[(x^2y^2)^2, z] \equiv 1$.

PROOF. We have $E \in \mathfrak{A}_2^3$, since $N \in \mathfrak{A}_2$ and $B \in \mathfrak{A}_2^2$. Since $x^2(E) \subset B$ and $x^2(B) \subset B'$, it follows that $(x^2y^2)^2(E) \subset B'$. But the relations defining the action of G on E prove that N centralizes B' , and thus that B' lies in the center of E . This concludes the proof.

§3. Free groups of the variety $\mathfrak{A}$

Set $\mathfrak{A} = \mathfrak{A}_2^3 \cap [\mathfrak{A}_2^2, \mathfrak{G}]$, where $\mathfrak{G}$ is the trivial variety; in other words, $\mathfrak{A}$ is the subvariety of $\mathfrak{A}_2^3$ defined in it by the identity $[(x^2y^2)^2, z] \equiv 1$. We set $L = F_\infty(\mathfrak{A})$ and denote free generators of L by $\{z_1, z_2, \dots\}$. In this section we investigate the structure of the verbal subgroup $L^4 = x^4(L)$ of L . We note to begin with that L^4 is center-by-metabelian, and that L^4 lies in the center of L .

1°. LEMMA 3.1. For arbitrary $u, w \in L$ and $v_1, v_2, v_3 \in L^2$, the following relations hold:

- a) $v_1^4 = 1, [v_1, v_2]^2 = 1, [v_1^2, u] = 1$;
- b) $[v_1, v_2, v_3] = [v_1, v_3][v_2, v_3], [v_1, v_2^{\pm 1}] = [v_2, v_1]$;
- c) $[v_1, v_2]^u = [v_1, v_2]$;
- d) $[(v_1v_2)^u, v_1v_2] = [v_1^u, v_1][v_2^u, v_2]$;
- e) $[(w^2)^u, w^2] = 1$.

PROOF. Relations a)–c) follow immediately since L is center-by-metabelian, and obviously $L^2 \in \mathfrak{A}_2^2 \cap \mathfrak{A}_2$. We prove d):

$$[(v_1v_2)^u, v_1v_2] \stackrel{b)}{=} [v_1^u, v_1][v_2^u, v_2][v_1^u, v_2][v_2^u, v_1].$$

But $[v_1^u, v_2] \stackrel{c)}{=} [v_1^u, v_2]^u = [v_1[v_1, u^2], v_2^u] \stackrel{b), c)}{=} [v_1, v_2^u]$, whence the required equation follows.

We prove e):

$$\begin{aligned} [(w^2)^u, w^2] &= [w^2(w^2)^u, w^2] = [w^2u^{-1}w^2u, w^2] \\ &= [w^2u^{-1}wuw^{-1}wu^{-1}wu, w^2] = [(wu^{-1}wu)^{w^{-1}} \cdot wu^{-1}wu, w^2]. \end{aligned}$$

Since $wu^{-1}wu \in L^2$, in view of b) we get

$$[(w^2)^u, w^2] = [(wu^{-1}wu)^{w^{-1}}, w^2][wu^{-1}wu, w^2] \stackrel{c)}{=} [wu^{-1}wu, w^2]^2 \stackrel{a)}{=} 1.$$

This proves the lemma.

LEMMA 3.2. For arbitrary $u \in L$ and $v \in L^2$, we have $(uv)^4 = u^4$.

PROOF. Since $u^2, v^uv \in L^2$, it follows from 3.1c) that

$$(uv)^4 = (u^2v^uv)^2 = u^4(v^uv)^2[v^uv, u^2].$$

But $[v^uv, u^2] \stackrel{b)}{=} [v^u, u^2][v, u^2] \stackrel{c)}{=} [v, u^2]^2 \stackrel{a)}{=} 1$, while

$$(v^uv)^2 = (v^2)^u v^2[v^u, v] \stackrel{a)}{=} v^4[v^u, v] \stackrel{a)}{=} [v^u, v].$$

Set $v = v_1^2 \cdots v_n^2$ ($v_i \in L$); then by d) and e) we get

$$(v^u v)^2 = [v^u, v] = \prod_{i=1}^n [(v_i^2)^u, v_i^2] = 1.$$

This proves the lemma.

The fundamental property of L^4 is contained in the following assertion.

LEMMA 3.3. (1) *The following set is a basis for the elementary abelian 2-group L^4 :*

$$\left\{ (z_{i_1} \cdots z_{i_k})^4 \mid 1 \leq i_1 < \cdots < i_k; k = 1, 2, \dots \right\}.$$

(w) *For arbitrary $v_1, v_2 \in L$,*

$$v_1^4 = v_2^4 \Leftrightarrow v_1 \equiv v_2 \pmod{L^2}.$$

PROOF. (1) In view of Lemma 3.2, the indicated set generates L^4 . If it were not an independent set of generators, there would be a relation of the form

$$(z_{i_1} \cdots z_{i_n})^4 = (z_{\alpha_1} \cdots z_{\alpha_k})^4 \cdots (z_{\beta_1} \cdots z_{\beta_m})^4,$$

where the different brackets enclose different ordered sets of free generators of L .

Suppose that $H = F_\infty(\mathfrak{A}_2)$ is freely generated by a set $h_1, h_2, \dots$. As in §2, we construct the group $C = C(H, H, M)$, where the set M contains just the single element $h_{i_1} \cdots h_{i_n}$. We consider the mapping $\eta: z_{i_n} \rightarrow h_{i_n} b^1, z_i \rightarrow h_i$ ($i \neq i_n$). By 2.2, $C(H, H, M) \in L$; therefore this map can be extended to a homomorphism $\eta: L \rightarrow C$. By Lemma 2.1,

$$(z_{i_1} \cdots z_{i_n})^4 \eta = (h_{i_1} \cdots h_{i_n} b^1)^4 \eta = (b^1_*)^{\chi(h_{i_1} \cdots h_{i_n})} = b^1_* \neq 1,$$

while at the same time

$$(z_{\alpha_1} \cdots z_{\alpha_k})^4 \eta = \cdots = (z_{\beta_1} \cdots z_{\beta_m})^4 \eta = 1$$

(since $\chi(h_{\alpha_1} \cdots h_{\alpha_k}) = \cdots = \chi(h_{\beta_1} \cdots h_{\beta_m}) = 0$). This contradiction proves the independence of the required elements.

(2) follows from Lemma 3.2 and (1). This proves the lemma.

REMARK 3.4. In terms of marginal subgroups, Lemma 3.3 can be stated in the following form. *Let G be any free group of $\aleph$. Then $(x^4)^*(G) = G^2$, and the set of all values of x^4 on representatives on the cosets $\{G \setminus G^2\} \bmod G^2$ is a set of independent generators of the elementary abelian 2-groups G^4 .*

The expression of an arbitrary element of L^4 in terms of the basis indicated in Lemma 3.3 will be said to be *canonical*.

2°. We come to a study of the lattice of verbal subgroups of L lying in L^4 .

LEMMA 3.5. *There is a word $v(z_1, \dots, z_m) = v_n$ in m variables in L^4 that reduces to the identity on deletion of any of the variables occurring in it (i.e. $v(z_1, \dots, z_{i-1}, 1, z_{i+1}, \dots, z_m) = 1$ ($i = 1, \dots, m$)). The element $v_m(z_1, \dots, z_m) \neq 1$ defined by this property is unique in L^4 .*

PROOF. Let $\lambda = (i_1, \dots, i_k)$, where $1 \leq i_1 < \cdots < i_k$. We denote $\pi(\lambda) = \pi(i_1, \dots, i_k) = (z_{i_1} \cdots z_{i_k})^4$ and set

$$v_m(z_1, \dots, z_m) = \prod_{\langle i_1, \dots, i_k \rangle \in \Lambda_m} \pi(i_1, \dots, i_k),$$

where Λ_m is the set of all subsets of $\{1, \dots, m\}$. It can be verified immediately that the word so constructed satisfies the conditions of the lemma. Indeed, the elements $\pi(\lambda)$ lie in an elementary abelian 2-group; that is, they commute with each other and are of order 2. When an arbitrary generator z_i is deleted from v_m , the set of new factors $\pi(\lambda')$ thus obtained coincides exactly with the set of factors not undergoing change (i.e., not containing z_i); as a result, the identity is obtained on contraction. The inequality $v_m \neq 1$ follows from Lemma 3.3.

Suppose now that $v(z_1, \dots, z_m) \in L^4$ is a nonidentity word which becomes the identity on deletion of an arbitrary variable. We call k the length of the factor $\pi(i_1, \dots, i_k)$ and consider the canonical decomposition of $v(z)$: $v(z_1, \dots, z_m) = \prod_{\lambda} z(\lambda)$. By induction on $m - k$, where k is the cardinal of λ , we show that for each $\lambda \in \Lambda_m$ the factor $\pi(\lambda)$ is actually present in the indicated decomposition.

Suppose that $k = m$, when the presence of $\pi(1, \dots, m)$ follows from the fact that if $\pi(i_1, \dots, i_n)$ ($n \leq m$) is the factor of maximal length, then on deletion from $v(z)$ of all variables not occurring in $\{z_{i_1}, \dots, z_{i_n}\}$, only one factor of length n remains in the resulting word, namely $\pi(i_1, \dots, i_n)$; thus by Lemma 3.3 it is not the identity, which means that $\{i_1, \dots, i_n\} = \{1, \dots, m\}$.

Suppose we have proved that factors $\pi(\lambda)$ occur in the canonical decomposition of $v(z)$ for all $\lambda \in \Lambda_m$ of cardinality $\geq m - k + 1$. We delete from $v(z)$ any k variables $z_{i_1}, \dots, z_{i_k}$. Set $\{1, \dots, m\} \setminus \{i_1, \dots, i_k\} = \{j_1, \dots, j_{m-k}\} = \lambda_1$. It is easy to see that deletion of factors of lengths $m, m-1, \dots, m-k+1$ gives the factor $\pi(\lambda_1)$ with frequencies of 1, $C_k', \dots, C_k^{k-1}$ respectively. Thus the whole collected part gives $\pi(\lambda_1)$ to the power

$$1 + C_k^1 + \dots + C_k^{k-1} = 2^k - 1 \equiv 1 \pmod{2},$$

whence it follows that $\pi(\lambda_1)$ must be a factor of the uncollected part. Since $\{i_1, \dots, i_k\}$ is arbitrary, this concludes the inductive step and with it the proof of the lemma.

LEMMA 3.6. *Suppose that $u_1, \dots, u_m$ are words in variables $z_1, \dots, z_{m-1}$. Then*

$$v_m(u_1, \dots, u_m) = 1.$$

PROOF. By Lemma 3.3, the words u_i can be viewed modulo L^2 ; that is, we may assume that $u_i = z_{\alpha_i} \cdots z_{\beta_i}$, where $1 \leq \alpha_i < \dots < \beta_i \leq m-1$. From the usual type of arguments on dependence of elements in finite-dimensional vector spaces, we get that $u_1, \dots, u_m$ are dependent modulo L^2 . Renumbering the u_i , we may assume that this dependence has the form $u_1 \equiv u_2 \cdots u_k \pmod{L^2}$ ($k \leq m$). In this case the element $v_m(u_1, \dots, u_m)$ is the image of $v = v_m(z_2, \dots, z_k, z_2, \dots, z_m)$ under the endomorphism $z_i \rightarrow u_i$ ($i = 1, \dots, m$). The element v is a word in variables $z_2, \dots, z_m$, and it follows from this that, on deletion of any of them, the word becomes the identity. Thus, by 3.5, the inequality $v \neq 1$ would give that $v = v_{m-1}(z_2, \dots, z_m)$. But, using the explicit form of the $v = v_m(z_2, \dots, z_m, z_2, \dots, z_m)$, it is easy to calculate that the power of the basis element z_2^4 in the canonical decomposition v is two. It then follows from the proof of Lemma 3.5 that $v \neq v_{m-1}(z_2, \dots, z_m)$, which means that $v = 1$. This proves the lemma.

LEMMA 3.7. *For all $m \geq 1$, $v_m(L) \not\supseteq v_{m+1}(L)$.*

PROOF. The inequality follows from the impossibility of the inclusion $v_m(z_1, \dots, z_m) \in v_{m+1}(L)$, which contradicts the fact proved in 3.6 that every value of the word $v_{m+1}(z)$ in $\text{gp}(z_1, \dots, z_m)$ is the identity.

The inequality $v_m(L) \supset v_{m+1}(L)$ follows from the equations

$$v_{m+1}(z_1, \dots, z_{m+1}) = v_m(z_1 z_{m+1}, z_2, \dots, z_m) \\ \times v_m(z_1, \dots, z_m) v_m(z_{m+1}, z_2, \dots, z_m),$$

which follows because of Lemma 3.5 and the fact that the word on the right-hand side is not the identity (since its canonical decomposition contains just one factor equal to $(z_1 \cdots z_{m+1})^4$), whereas deletion of any of the variables $z_1, \dots, z_{m+1}$ from it gives the identity.

THEOREM 3.8. *The lattice of verbal subgroups of L contained in L^4 is an infinite strictly decreasing chain. By the same token the set of varieties lying in the interval $\mathfrak{L} \supset \mathfrak{L} \cap \mathfrak{B}_4$ forms an infinite strictly increasing chain under inclusion.*

PROOF. By assertion 33.45 of [8], every word is equivalent to a collection of words, each of which becomes identically the unit element when any one of its variables is deleted. Thus, it follows from 3.5 that every verbal subgroup of L lying in L^4 is the fully invariant closure of some set of elements of the form $v_m(z)$, which means by 3.7 that it is the normal closure of just one such element. A second application of Lemma 3.7 concludes the proof of the theorem.

3°. Using Lemma 3.3, it is easy to establish another interesting property of L .

PROPOSITION 3.9. *Let L_1 be any noncyclic free group of $\mathfrak{L}$ freely generated by the set $\{z_1, z_2, \dots\}$. Then the characteristic subgroup N generated in L_1 by the element $w = z_1^4 z_2^4$ is not the same as the fully invariant subgroup generated by the same element.*

PROOF. Without loss of generality we may assume that $L_1 = L$. Every element $v \in N$ has the form $v = v_1 \cdots v_n$, where the v_i are images of w under automorphisms of L . The subgroup N lies in L^4 , and since the number of basic factors participating in cancellation is always even, it clearly follows from Lemma 3.3 that the number of basic factors in the canonical decomposition of v is even. Therefore the element z_1^4 in the fully invariant closure of z_1^4 does not lie in N .

COROLLARY 3.10. *The verbal subgroup generated by $t_1^4 t_2^4$ in any nonabelian absolutely free group F does not coincide with the characteristic subgroup of F generated by the same element.*

§4. Free groups of the variety $\mathfrak{L}\mathfrak{R}$

In this section $\mathfrak{R}$ will be any nontrivial variety of groups other than the variety of all groups. We denote by ρ the verbal subgroup of F_∞ corresponding to $\mathfrak{R}$, and by S a free noncyclic group of arbitrary rank of $\mathfrak{L}\mathfrak{R}$; $A = S/x^2(\rho(S))$ is the free group of $\mathfrak{L}_2\mathfrak{R}$ of rank the same as that of S .

As is well known (see [8], 21.13), the subgroup $\rho(S)$ of S is a free group of L , and $\rho(S) = L$ in the case where the factor-group $S/\rho(S)$ is countably infinite. Let $\{s_1, s_2, \dots\}$ and $\{y_1, y_2, \dots\}$ be sets of free generators of S and A respectively. We shall assume that $s_i \rightarrow y_i$ under the natural epimorphism $S \rightarrow S/x^2(\rho(S))$. It is easy to see that this epimorphism induces an isomorphism $\rho(A) \cong \rho(S)/x^2(\rho(S))$. In what follows we shall use the fact that for arbitrary words $v_1(t)$ and $v_2(t)$ the equation $v_1(y) = v_2(y)$ in A is equivalent to the congruence $v_1(s) \equiv v_2(s) \pmod{x^2(\rho(S))}$ in S .

1°. The following assertion is an immediate consequence of Lemma 3.3 and the remarks just made.

LEMMA 4.1. (1) Let $\{u_\alpha\}$ be a set of coset representatives of the set of cosets $\rho(S) \setminus x^2\rho(S)$ modulo $x^2(\rho(S))$. Then the set of all elements $\{u_\alpha^4\}$ is a set of independent generators of the elementary abelian 2-group $x^4(\rho(S))$ lying in the center of $\rho(S)$.

(2) Let $v_1(t), v_2(t) \in \rho(F_\infty)$. Then the following equivalence holds:

$$v_1^4(s) = v_2^4(s) \text{ in } S \Leftrightarrow v_1(y) = v_2(y) \text{ in } A.$$

We need another information of Lemma 4.1.

LEMMA 4.2. Suppose that $v_1(s), \dots, v_n(s) \in \rho(S) \setminus x^2(\rho(S))$ and that $v_1^4(s) \cdots v_n^4(s) = 1$. Then $n = 2k$, and the v_i may be renumbered in such a way that $v_{2m-1}(y) = v_{2m}(y)$ ($m = 1, \dots, k$).

Suppose now that $\{w_i(t)\}_{i \in I}, \{v_j(t)\}_{j \in J} \subset \rho(F_\infty)$ are arbitrary sets of words. A fundamental property of S is stated in the following theorem, which is an immediate corollary of Lemma 4.1.

THEOREM 4.3.

$$\text{gp}\{w_i^4(s) | i \in I\} \supset \text{gp}\{v_j^4(s) | j \in J\} \Leftrightarrow \bigcup_{i \in I} w_i(y) \supset \bigcup_{j \in J} v_j(y).$$

Theorem 4.3 may be used as a basis for studying the lattices of normal, characteristic (see §§5 and 6 of [2]) and fully invariant subgroups of S lying in $x^4(\rho(S))$. In this paper we consider applications of the theorem in the case of verbal subgroups.

COROLLARY 4.4.

$$\prod_{i \in I} w_i^4(S) \supset \prod_{j \in J} v_j^4(S) \Leftrightarrow \bigcup_{i \in I} w_i[A] \supset \prod_{j \in J} v_j[A].$$

REMARK 4.5. The subgroup $\rho(A)$ can be considered as a polygon over the semigroup End A (in the sense of [19]). It then follows from 4.4 that the lattice of all subpolygons of this polygon is isomorphically embedded in the lattice of verbal subgroups of S lying in $x^4(\rho(S))$.

A second important property of the product $\mathfrak{L}\mathfrak{R}$ is stated in the following assertion.

LEMMA 4.6. Suppose that $\mathfrak{R}$ is finitely based. Then the product $\mathfrak{L}\mathfrak{R}$ has a finite basis for its identities.

PROOF. It follows from the definition of $\mathfrak{L}$ and from properties of product varieties (see [8], 21.23) that $\mathfrak{L}\mathfrak{R} = \mathfrak{A}_2^3\mathfrak{R} \cap [\mathfrak{A}_2^2\mathfrak{R}, \mathfrak{R}]$. The assertion now follows since the product of an abelian variety and a finitely based variety is finitely based, and also from elementary arguments concerning the fact that intersections and commutators of finitely based varieties are finitely based.

2°. We now digress just a little.

PROPOSITION 4.7. If $\mathfrak{R} \subseteq \mathfrak{R}_c$, then for any variety $\mathfrak{M} \subseteq \mathfrak{L}$ the finitely generated free groups of $\mathfrak{M}\mathfrak{R}$ are defined by a finite number of identities.

PROOF. By a result of McKay [20], $\mathfrak{M}$ is finitely based and is defined in $\mathfrak{L}$ by the identity $w(t_1, \dots, t_l) \equiv 1$, say. We may suppose that $\mathfrak{M}$ is nontrivial, and therefore that

$w(t) \in F_\infty^2$. By a theorem of Lyndon (see [8], 34.14), $\mathfrak{R}$ is finitely based, which means that there exists a word $r(T) \in \rho(F_\infty)$ such that $r(F_\infty) = \rho(F_\infty)$. A basis for the identities of $\mathfrak{L}\mathfrak{R}$ in this case has the following form (see [8], 34.22):

$$\{v(t) \equiv 1\} \cup \{w(r_1^{\varepsilon_1} \cdots r_n^{\varepsilon_n}, \dots, r_{(l-1)n+1}^{\varepsilon_{(l-1)n+1}} \cdots r_{ln}^{\varepsilon_{ln}}) \equiv 1\}_{n=1,2,\dots}.$$

Here the identity $v(t) \equiv 1$ defines the variety $\mathfrak{L}\mathfrak{R}$ (Lemma 4.6); the r_i are copies of r written in nonintersecting sets of variables; $\varepsilon_i = \pm 1$.

By Lemma 1 of [1] (see also [2], 1.11 and 1.12), for arbitrary $m \geq 1$ there is a word $v_m(t) \in \gamma_c(F_\infty)$ such that $v_m[F_m(\mathfrak{A}_2^2\mathfrak{R}_c)] = \gamma_c(F_m(\mathfrak{A}_2^2\mathfrak{R}_c))$. Roman'kov [21] has remarked (and this is easy to prove, using Waring's hypothesis) that there exists a word $v'_m(t) \in \rho(F_\infty)$ such that $v'_m[F_m(\mathfrak{R}_c)] = \rho(F_m(\mathfrak{R}_c))$. It follows that the word $v'_m v_m$, where now v'_m and v_m are written in nonintersecting sets of variables, lies in $\rho(F_\infty)$, and that $f[F_m(\mathfrak{A}_2^2\mathfrak{R})] = \rho(F_m(\mathfrak{A}_2^2\mathfrak{R}))$. Since $\rho(F_\infty) = r(F_\infty)$, for some n the word $f(t)$ is a value of $r_1^{\varepsilon_1} \cdots r_n^{\varepsilon_n}$ in F_∞ ($\varepsilon_i = \pm 1$). We denote by ρ_1 the verbal subgroup of F_∞ corresponding to the variety $\mathfrak{A}_2^2\mathfrak{R}$, we set

$$V_k = \left\{ w(r_1^{\varepsilon_1} \cdots r_k^{\varepsilon_k}, \dots, r_{(l-1)k+1}^{\varepsilon_{(l-1)k+1}} \cdots r_{lk}^{\varepsilon_{lk}}) \mid \varepsilon_1 = \pm 1, \dots, \varepsilon_{lk} = \pm 1 \right\},$$

and we suppose that $f_1, \dots, f_l$ are copies of f written in nonintersecting sets of variables. From what was said above, it follows that the identity $\bar{w} = w(f_1, \dots, f_l) \equiv 1$ is a corollary of the identity

$$h = w(r_1^{\varepsilon_1} \cdots r_n^{\varepsilon_n}, \dots, r_{(l-1)n+1}^{\varepsilon_{(l-1)n+1}} \cdots r_{ln}^{\varepsilon_{ln}}) \equiv 1.$$

On the other hand, the subgroup $(L^2)^2$ of $L = F_\infty$ lies in the center of L and has exponent 2, so that $(L^2)^2 \subset (x^2)^*(L)$. It follows that $\rho_1(F_m(\mathfrak{L}\mathfrak{R}))$ lies in the marginal subgroup $(x^2)^*(\rho(F_m(\mathfrak{L}\mathfrak{R})))$; thus, since $w(t) \in F_\infty^2$ and because of the choice of the word $f \in \rho(F_\infty)$, it follows that for arbitrary k

$$\bar{w}[F_m(\mathfrak{L}\mathfrak{R})] \supset \{g[F_m(\mathfrak{L}\mathfrak{R})] \mid g(t) \in V_k\}.$$

This proves that in $F_m(\mathfrak{L}\mathfrak{R})$ the system of identities $\{g(t) \equiv 1 \mid g(t) \in V_k, k = 1, 2, \dots\}$ is a consequence of $\bar{w}(t) \equiv 1$, and therefore a consequence of $h(t) \equiv 1$. But then $F_m(\mathfrak{L}\mathfrak{R})$ is defined by the two identities $v(t) \equiv 1$ and $h(t) \equiv 1$. This proves the proposition.

Proposition 4.7 has a curious corollary. We set $\mathfrak{L}_1 = \mathfrak{L} \cap \mathfrak{B}_4$.

COROLLARY 4.8. *For arbitrary numbers n and c such that $n \neq 1$ and $n \geq c \geq 1$, the group $F_n(\mathfrak{L}_1\mathfrak{R}_c)$ is defined by a finite number of identities but does not have a finite basis for its identities.*

PROOF. The first part of the assertion follows from 4.7. By 35.13 of [18], the variety $\mathfrak{L}_1\mathfrak{R}_c$ is generated by its c -generator groups (by its 2-generator groups if $c = 1$); that is, a basis for the identities of $F_n(\mathfrak{L}_1\mathfrak{R}_c)$ is a basis for those of $\mathfrak{L}_1\mathfrak{R}_c$. It was proved in [22] that $\mathfrak{B}_4\mathfrak{A}$ is not finitely based. The proof of this fact was based on an application of the construction expounded in §2. Therefore in fact what was proved is that $(\mathfrak{B}_4 \cap \mathfrak{L})\mathfrak{A} = \mathfrak{L}_1\mathfrak{A}$ is infinitely based. Using the technique of [23], the factor $\mathfrak{A}$ is easily replaced by $\mathfrak{R}_c$ (to do this the groups A_n in [22] must be replaced by analogous groups, of class $c + 1$). This proves the corollary.

REMARK 4.9. *Using stronger additional arguments, the variety $\mathfrak{R}_c$ in 4.8 can be replaced by any of its subvarieties of zero exponent.*

§5. Module interpretation of verbal subgroups and a problem of P. Hall

In this section we consider a certain scheme for applying Lemma 4.1, based on a module interpretation of verbal subgroups of S lying $x^4(\rho(S))$.

Take an arbitrary word $v(t) \in \rho(F_\infty)$. The verbal subgroup $M = v^*(S)$ is an elementary abelian 2-group and thus can be considered as a cyclic right module over the semigroup ring $K = \mathbf{Z}_2(\text{End } S)$ with operation $\circ$ defined like this: for $g \in M$ and $\varphi_1, \dots, \varphi_n \in \text{End } S$ we set

$$g \circ (\varphi_1 + \dots + \varphi_n) = g\varphi_1 \cdots g\varphi_n.$$

Here the submodules of M are in one-to-one correspondence with the verbal subgroups of S lying in M . As usual, $M \cong K/\text{Ann}_K(v^4(S))$.

Let Δ_K be the augmentation ideal of K . The following lemma is an immediate consequence of Lemmas 4.1 and 4.2.

LEMMA 5.1. *Every element $\alpha \in \text{Ann}_K(v^4(S))$ may be represented in the form $\alpha = \alpha_1 + \alpha_2$, where the elements α_1 and α_2 have the form*

$$\begin{aligned} \alpha_1 &= d_1 + \dots + d_{2n} \in \Delta_K & (d_i \in \text{End } S \setminus \text{Ann}_K(v^4(S))), \\ \alpha_2 &= e_1 + \dots + e_k & (e_j \in \text{End } S \cap \text{Ann}_K(v^4(S))). \end{aligned}$$

Quite often it is possible to choose a variety $\mathfrak{R}$ and a word $v(t)$ such that the set $Q = \text{End } S \setminus \text{Ann}_K(v^4(S))$ of endomorphisms not annihilating $v^4(S)$ is a semigroup and its complement $\text{End } S \setminus Q$ is an ideal in $\text{End } S$. In this case M can be considered in a natural way as a module over $\mathbf{Z}_2(Q)$, such that every $\mathbf{Z}_2(Q)$ -submodule of M will be a K -submodule; by 5.1, $\text{Ann}_{\mathbf{Z}_2(Q)}(v^4(S))$ lies in the augmentation ideal of $\mathbf{Z}_2(Q)$.

We shall show how to use such a situation in the following assertion.

PROPOSITION 5.2. *There exists a relatively free group that is solvable of length 5 but not residually finite.*

PROOF. We set $\mathfrak{R} = \mathfrak{N}_2$, $S = F_2(\mathfrak{N}_2) = \text{gp}(s_1, s_2)$, $v = [[t_2, t_1, t_1], [t_2, t_1]]$ and $M = v^4(S)$. We first establish the identity

$$\text{Ann}_K(v^4(S)) \cap \text{End } S = \{ \varphi \in \text{End } S \mid |\varphi| = 0 \};$$

that is, we establish the following equivalence for $\varphi \in \text{End } S$:

$$v^4(s)\varphi = 1 \Leftrightarrow |\varphi| = 0. \quad (5.1)$$

The condition $|\varphi| = 0$ means that $s_1\varphi$ and $s_2\varphi$ lie in a cyclic subgroup modulo S' . It follows that $[s_1\varphi, s_2\varphi] \in \gamma_3(S) = \rho(S)$, and thus

$$v(S)\varphi = [[s_2\varphi, s_1\varphi, s_1\varphi], [s_2\varphi, s_1\varphi]] \in (\gamma_3(S))' \subset x^2(\rho(S)).$$

But by Lemma 4.1, $v^4(s)\varphi = (v(s)\varphi)^4 = 1$.

If $|\varphi| \neq 0$, then the elements $s_1\varphi$ and $s_2\varphi$ generate a free abelian group of rank 2 modulo S' . By Lemmas 1.1 and 1.2, $s_1\varphi$ and $s_2\varphi$ freely generate an $\mathfrak{N}_2$ -free group modulo $\gamma_3(S)$. But then, by Lemma 1.3, $s_1\varphi$ and $s_2\varphi$ freely generate an $\mathfrak{N}_2$ -free subgroup of S . An elementary calculation using the Magnus-Shmel'kin embedding (see §6) shows that $v(y) \neq 1$ in $A = F_2(\mathfrak{N}_2)$. Therefore, by 4.1, $v^4(s) \neq 1$; and this means that $v^4(s)\varphi \neq 1$.

Equation (5.1) proves that M is in fact a module over the semigroup ring $P = \mathbf{Z}_2(Q)$ of the semigroup of all nondegenerate endomorphisms $Q = \{ \varphi \in \text{End } S \mid |\varphi| \neq 0 \}$, where

every P -submodule M is obviously a K -submodule and $\text{Ann}_p(v^4(s)) \cap Q = \emptyset$. It thus follows from 5.1 that $\text{Ann}_p(v^4(s))$ lies in the augmentation ideal Δ_Q of P . This allows one to extend the homomorphism $P \rightarrow P/\text{Ann}_p(v^4(s)) \cong M$ to a homomorphism $P \rightarrow P/\Delta_Q \cong \mathbb{Z}_2$. We consider the factor-group S_1 of S obtained by factoring out of S the fully invariant subgroup generated by all elements of the form $v^4(s)\varphi_1(v^4(s)\varphi_2)^{-1}$ ($\varphi_1, \varphi_2 \in Q$).

We prove that S_1 is the required group. Indeed, S_1 is a relatively free group freely generated by elements r_1 and r_2 , say. By construction $v^4(r_1, r_2) \neq 1$, $|v^4(S_1)| = 2$, and for every endomorphism $\varphi \in \text{End } S_1$ we have

$$v^4(r_1, r_2)\varphi = \begin{cases} v^4(r_1, r_2), & \text{if } |\varphi| \neq 0, \\ 1, & \text{if } |\varphi| = 0. \end{cases}$$

In particular, $v^4(r_1^n, r_2) = v^4(r_1, r_2)$ for arbitrary $n \geq 1$. Let $\psi: S_1 \rightarrow G$ be a homomorphism onto a finite group G with $|G| = n$. Then, by what was said above,

$$v^4(r_1, r_2)\psi = v^4(r_1^n, r_2)\psi = v^4((r_1\psi)^n, r_2\psi) = v^4(1, r_2\psi) = 1;$$

that is, the image of the element $v^4(r_1, r_2) \neq 1$ is trivial in every finite factor-group. This proves the proposition.

The following theorem, which disproves the conjecture of Philip Hall formulated in the Introduction, is proved in the same way. We set

$$v_p(t) = \left[[t_1, t_2, t_1 t_2, t_1^2 t_2, \dots, t_1^{p-1} t_2]^2, [t_1, t_2, t_1, t_2]^2 \right].$$

THEOREM 5.3. *For every prime p there exists a relatively free group G that is solvable of length 5 such that $|G: (v_p^4)^*(G)| = p^2$, but $|v_p^4(G)| = 2$.*

PROOF. We set $\mathfrak{R} = \mathfrak{A}\mathfrak{A}_p$, $S = F_2(\mathfrak{R}) = \text{gp}(s_1, s_2)$ and

$$M = v_p^4(S), \quad u_1(t) = [t_1, t_2, t_1 t_2, t_1^2 t_2, \dots, t_1^{p-1} t_2], \quad u_2(t) = [t_1, t_2, t_1, t_2].$$

Then $v_p = [u_1^2, u_2^2]$.

We show first that, for every $\varphi \in \text{End } S$,

$$v_p^4(s)\varphi = 1 \Leftrightarrow |\varphi| \equiv 0 \pmod{p}. \quad (5.2)$$

For, the condition $|\varphi| \equiv 0 \pmod{p}$ means that the elements $s_1\varphi$ and $s_2\varphi$ lie in one and the same cyclic subgroup modulo $S^p \cdot S'$. But then at least one of the elements $s_1\varphi, s_2\varphi, s_1\varphi \cdot s_2\varphi, \dots, (s_1\varphi)^{p-1} \cdot s_2\varphi$ lies in $S^p \cdot S'$. If it is one of the first two elements, then $u_2^2(s)\varphi \in x^2(\rho(S))$, so that $v_p(s)\varphi \in x^2(\rho(S))$, which means by 4.1 that $v_p^4(s)\varphi = (v_p(s)\varphi)^4 = 1$. If one of the elements of the form $(s_1\varphi)^k \cdot s_2\varphi$ lies in $S^p \cdot S'$, then $u_1^2(s)\varphi \in x^2(\rho(S))$, and thus $v_p^4(s)\varphi = 1$.

Suppose now that $|\varphi| \not\equiv 0 \pmod{p}$. Then the elements $s_1\varphi$ and $s_2\varphi$ $\mathfrak{A}_p$ -freely generate a subgroup modulo $S^p S'$, and thus in view of 1.3 they generate a subgroup modulo $\rho(S)$ $\mathfrak{A}_p$ -freely. We show that the elements $u_1^2(s)\varphi$ and $u_2^2(s)\varphi$ $\mathfrak{A}$ -freely generate a subgroup modulo $\rho(S)$. By what was said above, it is enough to prove this for $\varphi = 1$. We use the Magnus embedding (see [8], 36.12)

$$F_2/(F_2^p F_2')' \rightarrow G \text{ wr } H,$$

where

$$G = \langle g_1, g_2 | [g_1, g_2] = 1 \rangle, \quad H = \langle X, Y | [X, Y] = X^p = Y^p = 1 \rangle.$$

Using the module description for the base group of the wreath product, which is a free right module on $\{g_1, g_2\}$ over $\mathbf{Z}(H)$, we get

$$u_1^2(Xg_1, Yg_2) = 2(g_1(Y-1) - g_2(X-1))(XY-1) \cdots (X^{p-1}Y-1),$$

$$u_2^2(Xg_1, Yg_2) = 2(g_1(Y-1) - g_2(X-1))(X-1)(Y-1).$$

If the elements $u_1(s)$ and $u_2(s)$ were independent modulo $\rho(S)$, for some $m, n \in \mathbf{Z} \setminus \langle 0 \rangle$, we would have $u_2^{2m}(s) \equiv u_2^{2n}(s) \pmod{\rho(S)}$, which on transition to the wreath product would give the following equality in the first coordinate subgroup:

$$2m(Y-1)(XY-1) \cdots (X^{p-1}Y-1) = 2n(Y-1)^2(X-1).$$

Setting $X = 1$, we get that $(Y-1)^p = 0$ in $\mathbf{Z}(\text{gp}(Y))$. But

$$(Y-1)^p = \sum_{k=1}^{p-1} (-1)^k \binom{p}{k} Y^k \neq 0.$$

This contradiction proves that the elements $u_1^2(s)\varphi$ and $u_2^2(s)\varphi$ $\mathfrak{A}$ -freely generate a subgroup modulo $\rho(S)$. But these elements lie in $S^p \cdot S'$, which is a free group of $\mathfrak{A}$. Since $(S^p S')' = \rho(S)$, by 1.3 the elements $u_1^2(s)\varphi$ and $u_2^2(s)\varphi$ $\mathfrak{A}$ -freely generate a subgroup of S , whence it follows that $[u_1^2(s)\varphi, u_2^2(s)\varphi] \notin x^2(\rho(S))$, which means that $v_p^4(s)\varphi = [u_1^2(s)\varphi, u_2^2(s)\varphi]^4 \neq 1$. This concludes the proof of (5.2).

We set $Q_p = \{\varphi \in \text{End } S \mid |\varphi| \not\equiv 0 \pmod{p}\}$ and $P = \mathbf{Z}_2(Q_p)$. It follows from (5.2) that M is in fact a module over P such that every P -submodule is a K -submodule; and it follows from Lemma 5.1 that $\text{Ann}_p(v_p^4(s)) \subset \Delta_{Q_p}$. Just as in the proof of Proposition 5.2, this lets us extend the homomorphism $P \rightarrow P/\text{Ann}_p(v_p^4(s)) \cong M$ to a homomorphism $P \rightarrow P/\Delta_{Q_p} \cong \mathbf{Z}_2$. We shall prove that the factor-group of G by the fully invariant subgroup generated by all elements of the form $v_p^4(s)\varphi_1(v_p^4(s)\varphi_2)^{-1}$ ($\varphi_1, \varphi_2 \in Q_p$) is the one we want.

Let f_1 and f_2 be free generators of G . By construction $v_p^4(f_1, f_2) \neq 1$, $|v_p^4(G)| = 2$, and, for every $\varphi \in \text{End } G$,

$$v_p^4(f_1, f_2)\varphi = \begin{cases} v_p^4(f_1, f_2), & \text{if } |\varphi| \not\equiv 0 \pmod{p}, \\ 1, & \text{if } |\varphi| \equiv 0 \pmod{p}. \end{cases} \quad (5.3)$$

But this equation means exactly that $(v_p^4)^*(G) = G^p G'$, and thus that $|G: (v_p^4)^*(G)| = p^2$. In fact the substitution $v_p^4(r_1 d_1, r_2 d_2)$, where $r_i \in G$ and $d_i \in G^p G'$, is equivalent to applying the endomorphism φ defined by the equations $f_i \varphi = r_i d_i$ ($i = 1, 2$). But if we define the endomorphism φ' by $\varphi': f_i \rightarrow r_i$, then clearly $|\varphi| \equiv |\varphi'| \pmod{p}$, and consequently $v_p^4(r_1 d_1, r_2 d_2) = v_p^4(r_1, r_2)$. This proves that $(v_p^4)^*(G) \supset G^p G'$. We choose now an arbitrary element $r \in G \setminus G^p G'$; then r has the form $r = f_1^m f_2^n h$, where $h \in G^p G'$ and $(m, n) \not\equiv (0, 0) \pmod{p}$. Suppose for example that $m \not\equiv 0 \pmod{p}$. Then

$$\begin{vmatrix} m & n \\ 0 & 1 \end{vmatrix} = m \not\equiv 0 \pmod{p},$$

so that, by (5.3), $v_p^4(r, f_2) \neq 1$. But $v_p^4(r \cdot r^{-1}, f_2) = v_p^4(1, f_2) = 1$. This means that $(v_p^4)^*(G) \subset G^p G'$, and concludes the proof of the theorem.

§6. Infinite direct products of verbal subgroups

As we proved in 3.8, the lattice of all verbal subgroups of $L = F_\infty(\mathfrak{N})$ lying in L^4 is an infinite strictly decreasing chain. In this section we consider how the situation changes under the natural embedding of L in $S = F_\infty(\mathfrak{N}\mathfrak{R})$ (via the isomorphism $L \cong \rho(S)$), and show that when $\mathfrak{R} = \mathfrak{R}_c$ ($c \geq 9$), there exist infinite direct products of verbal subgroups of S lying in $x^4(\rho(S)) = x^4(\gamma_{c+1}(S))$. By Theorem 4.3, it is enough here to indicate an infinite system of words $\{w_i(t)\}_{i \in \mathbb{Z}} \subset \gamma_{c+1}(F_\infty)$ such that for $i \neq j$ we have $w_i[A] \cap w_j[A] = 1$ in $A = F_\infty(\mathfrak{A}_2\mathfrak{R}_c)$, but $w_i[A] \neq 1$ for $i \in \mathbb{Z}$. As will be proved in turn, it will be enough to construct a system of words $\{g_i(t)\}_{i \in \mathbb{Z}} \subset \gamma_c(F_\infty)$ such that $g_i[F_\infty(\mathfrak{R}_c)] \neq 1$ for all $i \in \mathbb{Z}$, whereas for $i \neq j$ the equation $v_i^{m_1} = v_j^{m_2}$, where $v_k \in [F_\infty(\mathfrak{R}_c)]$ and $m_1, m_2 \in \mathbb{Z} \setminus \{0\}$, implies that $v_i = v_j = 1$.

1°. We construct such a system of words $\{g_i(t)\}$. The proof will rest on commutator calculations in a free nilpotent group.

Let $l_0 \geq 1$ be an integer; we set $c = 2l_0 + 7$ and define four commutators, the first of which is of weight $c + 1$, and the other three of weight c :

$$\begin{aligned} u_0(t) &= \left[t_2, \underbrace{t_1, \dots, t_1}_c \right], \\ u_1(t) &= \left[[t_2, t_1, t_2], [t_2, t_1, t_1], [t_2, t_1, t_1], \underbrace{[t_2, t_1], \dots, [t_2, t_1]}_{l_0-1} \right], \\ u_2(t) &= \left[[t_2, t_1, t_1, t_2], \underbrace{[t_2, t_1], \dots, [t_2, t_1]}_{l_0}, [t_2, t_1, t_1] \right], \\ u_3(t) &= \left[[t_2, t_1, t_2], \underbrace{[t_2, t_1], \dots, [t_2, t_1]}_{l_0}, [t_2, t_1, t_1, t_2] \right]. \end{aligned}$$

We fix notation: $g_i(t) = u_i'(t)u_2(t)u_3(t)$ and $w_i(t) = [u_0(t), g_i(t)]$. In this section we set also $\mathfrak{R} = \mathfrak{R}_c$ and $\rho = \gamma_{i+1}$, and let $\{x_1, x_2, \dots\}$ be a set of free generators of $R = F_\infty(\mathfrak{R}_c) = S/\gamma_{c+1}(S)$, $R_n = \text{gp}(x_1, \dots, x_n)$, and $\{y_1, y_2, \dots\}$ a set of free generators of $A = F_\infty(\mathfrak{A}_2\mathfrak{R}_c) = S/x^2(\gamma_{c+1}(S))$.

LEMMA 6.1. *Take $v(x_1, x_2) \in \gamma_c(R) \setminus \{1\}$ and $\varphi \in \text{End } R$. Then $v(x)\varphi = 1 \Leftrightarrow \text{gp}(x_1\varphi, x_2\varphi)$ is cyclic modulo R' .*

PROOF. The word $v(x)$ can be represented as a product of commutators of weight c in x_1 and x_2 . Thus, by virtue of the distributivity in each place of commutators of weight c in R , the dependence of the elements $x_1\varphi$ and $x_2\varphi$ modulo R' gives $v(x)\varphi = v(x_1\varphi, x_2\varphi) = 1$. Conversely, suppose that $x_1\varphi$ and $x_2\varphi$ are independent modulo R' . Then, by 1.1 and 1.2, they freely generate an $\mathfrak{R}_c$ -free subgroup of R , so that $v(x)\varphi \neq 1$. This proves the lemma.

We fix notation for the following basic commutators in x_1 and x_2 :

$$\begin{aligned} b &= [x_2, x_1], \quad c_1 = [x_2, x_1, x_1], \quad c_2 = [x_2, x_1, x_2], \\ d_1 &= [x_2, x_1, x_1, x_1], \quad d_2 = [x_2, x_1, x_1, x_2], \quad d_3 = [x_2, x_1, x_2, x_2]. \end{aligned}$$

The following assertion is a strengthening of Lemma 4 of [24], and describes an important property of the word $u_2(x)u_3(x)$ in R_2 .

LEMMA 6.2. *For arbitrary endomorphisms $\varphi_1, \varphi_2 \in \text{End } R_2$,*

$$(u_2(x)u_3(x))\varphi_1 = (u_2(x)u_3(x))\varphi_2 \neq 1 \Leftrightarrow |\varphi_1| \cdot |\varphi_2| \neq 0 \quad \text{and} \quad (\varphi_1) = (\varphi_2).$$

PROOF. The commutators $u_2(x)$ and $u_3(x)$ are basic; therefore $u_2(x) \neq 1$, $u_3(x) \neq 1$ and $u_2(x)u_3(x) \neq 1$. By 6.1, we may assume that $|\varphi_1| \cdot |\varphi_2| \neq 0$. We give the proof for $c = 9$, since all arguments in the general case are the same.

Let $\varphi \in \text{End } R_2$ be any endomorphism. Clearly, the action of φ on $u_2(x)u_3(x) \in \gamma_c(R_2)$ is defined uniquely by the matrix (φ) ; that is, by the action of $\varphi \bmod R'_2$. To prove the lemma it suffices to prove that when $|\varphi| \neq 0$, the matrix (φ) is uniquely determined by $(u_2(x)u_3(x))\varphi$.

We compute the decomposition of $(u_2(x)u_3(x))\varphi$ as a product of basic commutators of weight c . Suppose that

$$\left. \begin{aligned} x_1\varphi &\equiv x_1^k x_2^l \\ x_2\varphi &\equiv x_1^m x_2^n \end{aligned} \right\} \pmod{R'_2}, \quad \text{i.e. } (\varphi) = \begin{pmatrix} k & l \\ m & n \end{pmatrix}.$$

We can write $u_2(x) = [d_2, b, c_1]$ and $u_3(x) = [c_2, b, d_2]$. We let $q = |\varphi| = kn - ml$; then direct calculation proves that modulo the corresponding terms of the lower central series of R_2 we have

$$b\varphi \equiv b^q, \quad c_1\varphi \equiv c_1^q c_2^l, \quad c_2\varphi \equiv c_1^m c_2^n, \quad d_2\varphi \equiv d_1^{qkm} d_2^{q(kn+lm)} d_3^{qln}.$$

Using these equations, we get decompositions as products of different (and thus independent—see [8], 31.61) basic commutators

$$\begin{aligned} u_2(x)\varphi &= [d_1, b, c_1]^{q^3 k^2 m} [d_1, b, c_2]^{q^3 k l m} [d_2, b, c_1]^{q^3 k(kn+lm)} \\ &\quad \times [d_2, b, c_2]^{q^3 l(kn+lm)} [d_3, b, c_1]^{q^3 k l n} [d_3, b, c_2]^{q^3 l^2 n}, \\ u_3(x)\varphi &= [c_1, b, d_1]^{q^3 k m^2} [c_1, b, d_2]^{q^3 m(kn+lm)} [c_1, b, d_3]^{q^3 l m n} \\ &\quad \times [c_2, b, d_1]^{q^3 k m n} [c_2, b, d_2]^{q^3 n(kn+lm)} [c_2, b, d_3]^{q^3 l n^2}. \end{aligned}$$

Denoting by α_{ij} and β_{ij} respectively the powers of the basic commutators $[d_i, b, c_j]$ and $[c_i, b, d_j]$ in these equations, we get

$$\begin{aligned} q^c &= q^9 = q^6 \cdot q^3 = q^6 (n^3 k^3 - 3n^2 k^2 m l + 3n k m^2 l^2 - m^3 l^3) \\ &= (\beta_{22} - \beta_{13})(\alpha_{21} - \alpha_{12}) - 3\alpha_{11}\beta_{23} + 3\alpha_{32}\beta_{11} - (\alpha_{22} - \alpha_{31})(\beta_{12} - \beta_{21}). \end{aligned} \quad (6.1)$$

This proves that the number $q = |\varphi| \neq 0$ is uniquely determined by $(u_2(x)u_3(x))\varphi$. Analogously we get

$$\begin{aligned} kq^{(c-1)/2} &= kq^4 = (k^2 n - k m l)q^3 = \alpha_{21} - 2\alpha_{12}, \\ lq^{(c-1)/2} &= lq^4 = (l n k - m l^3)q^3 = 2\alpha_{31} - \alpha_{22}, \\ mq^{(c-1)/2} &= mq^4 = (m k n - m^2 l)q^3 = 2\beta_{21} - \beta_{12}, \\ nq^{(c-1)/2} &= nq^4 = (n^2 k - m l n)q^3 = \beta_{22} - 2\beta_{13}. \end{aligned}$$

Finally,

$$\left. \begin{aligned} k &= q^{(1-c)/2}(\alpha_{21} - 2\alpha_{12}), & l &= q^{(1-c)/2}(2\alpha_{31} - \alpha_{22}), \\ m &= q^{(1-c)/2}(2\beta_{21} - \beta_{12}), & n &= q^{(1-c)/2}(\beta_{22} - 2\beta_{13}). \end{aligned} \right\} \quad (6.2)$$

Thus, the matrix (φ) is uniquely determined by $(u_2(x)u_3(x))\varphi$, and the lemma is proved.

LEMMA 6.3. *For arbitrary endomorphisms $\varphi_1, \varphi_2 \in \text{End } R$, the following conditions are equivalent:*

- (I) $(u_2(x)u_3(x))\varphi_1 = (u_2(x)u_3(x))\varphi_2 \neq 1$;
- (II) $x_i\varphi_1 \equiv x_i\varphi_2 \pmod{R'}$, and the group $G_i = \text{gp}(x_1\varphi_i, x_2\varphi_i)$ is not cyclic modulo R' ($i = 1, 2$).

PROOF. (I) $\Rightarrow$ (II). Without loss of generality, R can be replaced by R_n . Suppose that the matrices (φ_1) and (φ_2) have the form

$$(\varphi_1) = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \dots & \dots & \dots \end{pmatrix}, \quad (\varphi_2) = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ b_{21} & \cdots & b_{2n} \\ \dots & \dots & \dots \end{pmatrix}.$$

We denote by A_i (by B_i) the i th row of (φ_1) (of (φ_2)),

$$a_i = \begin{pmatrix} a_{1i} \\ a_{2i} \end{pmatrix}, \quad b_i = \begin{pmatrix} b_{1i} \\ b_{2i} \end{pmatrix},$$

and we define an endomorphism $\delta_{kl} \in \text{End } R_n$ by the relations $x_k\delta_{kl} = x_1$, $x_l\delta_{kl} = x_2$, and $x_j\delta_{kl} = 1$ for $j \notin \{k, l\}$.

By 6.1, G_i is not cyclic modulo R'_n ; that is, $\text{rank}(A_1, A_2) = \text{rank}(B_1, B_2) = 2$. We consider the k th columns a_k and b_k . Two cases are possible: $a_k = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $a_k \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. In the first case, applying δ_{kl} to the equation in the conditions of the lemma, and using Lemma 6.1, we get that $\text{rank}(a_k, a_l) = \text{rank}(b_k, b_l) \leq 1$ ($1 \leq l \leq n$). It follows that $b_k = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = a_k$ (otherwise $\text{rank}(B_1, B_2) < 2$). Applying δ_{kl} and using the fact that $R_2\varphi_i\delta_{kl} \subseteq R_2$, from 6.2 we get that $(a_k, a_l) = (b_k, b_l)$, which means that $a_k = b_k$.

(II) $\Rightarrow$ (I). This follows from 6.1 and the fact that $u_2(x)u_3(x) \in R_2$.

2°. LEMMA 6.4. *For arbitrary endomorphisms $\varphi_1, \varphi_2 \in \text{End } R$ and nonzero integers m_1 and m_2 , the following two conditions are equivalent:*

- (I) $((u_2(x)u_3(x))\varphi_1)^{m_1} = ((u_2(x)u_3(x))\varphi_2)^{m_2} \neq 1$;
- (II) $m_1^{1/c}A_i = m_2^{1/c}B_i$, and the groups $G_i = \text{gp}(x_1\varphi_i, x_2\varphi_i)$ are not cyclic modulo R' ($i = 1, 2$). (Here A_i and B_i are the i th rows of (φ_1) and (φ_2) respectively.)

PROOF. Since R is nilpotent and torsion-free, the extraction of roots is unique (this follows from [8], 31.13 and 31.41, for example). Therefore we may assume that $(m_1, m_2) = 1$ in (I). If the groups G_i are not cyclic modulo R' , then $\text{rank}(A_1, A_2) = \text{rank}(B_1, B_2) = 2$, and this means that the rows A_1 and B_1 are nonzero. In this case the equation $m_1^{1/c}A_1 = m_2^{1/c}A_2$ and the condition $(m_1, m_2) = 1$ allow us to conclude that the numbers $n_1 = m_1^{1/c}$ and $n_2 = m_2^{1/c}$ are integers. But then, for any element $v(x) \in \gamma_c(R)$ we have $v(x)^{m_i} = v(x)\psi_i$, where the endomorphism $\psi_i \in \text{End } R$ is defined by

$$x_k\psi_i = x_k^n \quad (i = 1, 2; k = 1, 2, \dots).$$

Therefore from $n_1 A_i = n_2 B_i$ ($i = 1, 2$) there follows

$$\begin{aligned} ((u_2(x)u_3(x))\varphi_1)^{m_1} &= (u_2(x)u_3(x))\varphi_1\psi_1 \\ &= (u_2(x)u_3(x))\varphi_2\psi_2 = ((u_2(x)u_3(x))\varphi_2)^{m_2}, \end{aligned}$$

whose parts are not the identity, by 6.1. This proves that (II) $\Rightarrow$ (I).

Conversely, suppose that (I) holds. Then, by Lemma 6.1, the groups G_i are not cyclic mod R' ; that is, $\text{rank}(A_1, A_2) = \text{rank}(B_1, B_2) = 2$. We assume first that $(u_2(x)u_3(x))\varphi_i \in R_2$. Then we may assume that $\varphi_1, \varphi_2 \in \text{End } R_2$. As with φ in the proof of Lemma 6.2 we denote by $\alpha_{ij}^{(1)}, \beta_{ij}^{(1)}$ and $\alpha_{ij}^{(2)}, \beta_{ij}^{(2)}$ the powers of the basic commutators in the decompositions of $u_2(x)\varphi_1, u_3(x)\varphi_1$ and $u_2(x)\varphi_2, u_3(x)\varphi_2$ respectively. Equation (1) then becomes a system of equations $\{m_1\alpha_{ij}^{(1)} = m_2\alpha_{ij}^{(2)}, m_1\beta_{ij}^{(1)} = m_2\beta_{ij}^{(2)}\}$, where the inequality in (I) is equivalent to the condition that $|\varphi_1||\varphi_2| \neq 0$. Calculating the expression for $|\varphi_i|$ by analogy with (6.1), we get $m_1^2|\varphi_1|^c = m_2^2|\varphi_2|^c$. It follows from the analogues of (6.2) for the elements of

$$(\varphi_i) = \begin{pmatrix} k^{(i)} & l^{(i)} \\ m^{(i)} & n^{(i)} \end{pmatrix}$$

and from the previous discussions that

$$\begin{aligned} m_1^{1/c}k^{(1)} &= n_1 \frac{\alpha_{21}^{(1)} - 2\alpha_{12}^{(1)}}{|\varphi_1|^{(c-1)/2}} = n_1 \frac{\alpha_{21}^{(1)} - 2\alpha_{12}^{(1)}}{n_2^{c-1}n_1^{1-c}|\varphi_2|^{(c-1)/2}} \\ &= n_1 \frac{m_1^{-1}m_2(\alpha_{21}^{(2)} - 2\alpha_{12}^{(2)})}{n_2^{c-1}n_1^{1-c}|\varphi_2|^{(c-1)/2}} = m_2^{1/c}k^{(2)}, \end{aligned}$$

and so on. As a result we get the required equation $m_1^{1/c}(\varphi_1) = m_2^{1/c}(\varphi_2)$. The general case reduces to the one just analyzed on application of the projection endomorphisms δ_{kl} just as in the proof of Lemma 6.3. This proves the lemma.

As is easy to see, $u_1(x) \neq 1$, and thus $u_1(R) \neq 1$.

LEMMA 6.5. $u_1(R) \cap u_2(R) \cdot u_3(R) = 1$.

PROOF. Using the distributivity in every place of commutators of weight c in R , we can represent every value of $u_1(t)$ in R as a product of commutators in the x -alphabet, each of which contains within it three left-normalized commutators of weight 3 and $l_0 - 1$ left-normalized commutators of weight 2. To decompose such a commutator as a product of basic commutators, one uses only distributivity, the Jacobi identity and the identity $[u, v] = [v, u]^{-1}$. Therefore the number and weight of left-normalized commutators occurring in the basic commutators thus obtained does not change. We have proved that every element of $u_1(R)$ is representable as a product of basic commutators of weight c , each of which involves three left-normalized commutators of weight 3 and $(l_0 - 1)$ left-normalized commutators of weight 2. Analogously, every element of $u_2(R)u_3(R)$ is a product of basic commutators involving only one left-normalized commutator of weight 3 (the remaining left-normalized commutators are of weight 2 or 4). The lemma now follows from the independence of basic commutators.

LEMMA 6.6. Let $\varphi_1, \varphi_2 \in \text{End } R$; let n_1, n_2, m_1 and m_2 be arbitrary integers with $n_1 \neq n_2$. Then the equation $(g_{n_1}(x)\varphi_1)^{m_1} = (g_{n_2}(x)\varphi_2)^{m_2}$ implies that both sides are the identity.

PROOF. From Lemma 6.5 we have

$$(u_1(x)\varphi_1)^{n_1m_1} = (u_1(x)\varphi_2)^{n_2m_2}, \quad (6.3)$$

$$((u_2(x)u_3(x))\varphi_1)^{m_1} = ((u_2(x)u_3(x))\varphi_2)^{m_2}. \quad (6.4)$$

Assume that $(g_{n_1}(x)\varphi_1)^{m_1} = (g_{n_2}(x)\varphi)^{m_2} \neq 1$. Then $m_1m_2 \neq 0$ and the groups $\text{gp}(x_1\varphi_i, x_2\varphi_2)$ ($i = 1, 2$) are not cyclic modulo R' . In this case Lemma 6.1 asserts that both sides of (6.3) and (6.4) are not the identity. By Lemma 6.4, $m_1^{1/c}A_i = m_2^{1/c}B_i$ ($i = 1, 2$), whence it follows that $(u_1(x)\varphi_1)^{m_1} = (u_1(x)\varphi_2)^{m_2} \neq 1$ (since the action of the φ_i on $u_i(x_1, x_2)$ is uniquely determined by the first two rows of (φ_i) , and the numbers $m_i^{1/c}$ can be considered to be integral—see the proof of Lemma 6.4). But then it follows from (6.3) that

$$\begin{aligned} 1 &= (u_1(x)\varphi_1)^{n_1m_1}(u_1(x)\varphi_2)^{-n_2m_2} \\ &= (u_1(x)\varphi_1)^{n_1m_1}((u_1(x)\varphi_1)^{m_1})^{-n_2} = (u_1(x)\varphi_1)^{m_1(n_1-n_2)}, \end{aligned}$$

which contradicts the fact that R is torsion-free. This proves the lemma.

3°. We define the Magnus-Shmel'kin embedding (see [8], 36.12 and 22.48) $\mu: A \rightarrow H \text{ wr } R$ of $A = F_\infty(\mathfrak{A}_2\mathfrak{R})$ in the restricted wreath product of $H = F_\infty(\mathfrak{A}_2)$ and R . If $\{h_1, h_2, \dots\}$ is a set of free generators of H , then the monomorphism μ is defined by the relations $y_i\mu = x_ih_i$ ($i = 1, 2, \dots$). In module form, the base group $M = H^R$ of the wreath product can be considered as a free right module over $\mathbf{Z}_2(R)$ with basis $\{h_i\}$. For any subgroup $G \subset R$, we denote by Δ_G the two-sided ideal of $\mathbf{Z}_2(R)$ generated by all elements of the form $(g - 1)$, $g \in G$.

LEMMA 6.7. *Suppose that the endomorphism $\varphi \in \text{End } A$ is such that $\text{gp}(y_1\varphi, y_2\varphi)$ is not cyclic modulo A' . Then $u_0(y_1, y_2)\varphi\mu \notin M \cdot \Delta_{R'}$.*

PROOF. The natural epimorphism $R \rightarrow R/R'$ induces an epimorphism $\kappa: H \text{ wr } R \rightarrow H \text{ wr } (R/R')$ of wreath products such that $M \cap \text{Ker } \kappa = M \cdot \Delta_{R'}$ (see [8], 26.21). Obviously, the product map $\mu\kappa$ is the product of two homomorphisms, the first of which is the natural epimorphism $\psi: A \rightarrow B \cong F_\infty(\mathfrak{A}_2\mathfrak{A})$, and the second is the Magnus-Shmel'kin embedding of B in $H \text{ wr } (R/R')$.

Let φ' be the endomorphism of B induced by φ under ψ . Then $u_0(y)\varphi\psi = u_0(y)\psi\varphi'$. The elements $y_1\psi\varphi'$ and $y_2\psi\varphi'$ of B are independent modulo B' , so that by 1.3 they $\mathfrak{A}_2\mathfrak{A}$ -freely generate a subgroup of B , and thus $u_0(y)\psi\varphi' \neq 1$. But then $u_0(y)\varphi\mu\kappa \neq 1$, which means that $u_0(y)\varphi\mu \notin M \cdot \Delta_{R'}$. This proves the lemma.

LEMMA 6.8. *Let there be given words $v_1(t), v_2(t) \in \gamma_c(F_\infty)$ and endomorphisms $\varphi_1, \varphi_2 \in \text{End } A$ such that at least one of the groups $G_i = \text{gp}(y_1\varphi_i, y_2\varphi_i)$ ($i = 1, 2$) is not cyclic modulo A' , and also*

$$[u_0(y), v_1(y)]\varphi_1 = [u_0(y), v_2(y)]\varphi_2.$$

Then $G = \text{gp}(v_1(y)\varphi_1, v_2(y)\varphi_2)$ is cyclic modulo $\gamma_{c+1}(A)$.

PROOF. We use the Magnus-Shmel'kin embedding. On transition to the wreath product, the equation in the hypothesis becomes the following equation in the free module M :

$$f_1(v_1(x)\varphi'_1 - 1) = f_2(v_2(x)\varphi'_2 - 1),$$

where $f_i = u_0(y)\varphi_i\mu \in M$ and φ'_i denotes the endomorphism of R induced by φ_i under the natural epimorphism $A \rightarrow A/\gamma_{c+1}(A) \cong R$ (we have also used the obvious relation $v_i(y)\varphi_i\mu \equiv v_i(x)\varphi'_i \pmod{M}$).

Suppose, for example, that G_2 is not cyclic modulo A' . We set $v_1(x)\varphi'_1 = d^m$, where $d \in \gamma_c(R)$ is not a proper power in R . We set $D = \text{gp}(d)$, and then $f_2(v_2(x)\varphi'_2 - 1) \in M \cdot \Delta_D$. The factor-group R/D is torsion-free and nilpotent, so that the ring $\mathbb{Z}_2(R)/\Delta_D = \mathbb{Z}_2(R)/\Delta$ does not have zero-divisors. But it follows from this that $v_2(x)\varphi'_2 - 1 \in \Delta_D$. This is equivalent to the fact that $v_2(x)\varphi'_2 \in D$, which means that G is cyclic modulo $\gamma_{c+1}(A)$. This proves the lemma.

We can now come to a consideration of the sets of values of words w_i in A . We note first that $w_i(y) \neq 1$, which means that $w_i[A] \neq 1$ for every $i \in \mathbb{Z}$. It is easy to check that R is also torsion-free, using the Magnus-Shmel'kin embedding.

LEMMA 6.9. *For arbitrary integers $i \neq j$, $w_i[A] \cap w_j[A] = 1$*

PROOF. Every value of the word $w_i(t)$ in A can be represented in the form $w_i(y)\varphi = [u_0(y), g_i(y)]\varphi$, where $\varphi \in \text{End } A$. If $\text{gp}(y_1\varphi, y_2\varphi)$ is cyclic modulo A' , 6.1 gives that $g_i(y)\varphi \in \gamma_{c+1}(A)$, which means that $w_i(y)\varphi = 1$. Thus it follows from the relation $w_i(y)\varphi_1 = w_j(y)\varphi_2 \neq 1$ that the groups $G_i = \text{gp}(y_1\varphi_i, y_2\varphi_i)$ ($i = 1, 2$) are not cyclic modulo A' (here $\varphi_i \in \text{End } A$). But then it follows from Lemma 6.8 that $G = \text{gp}(g_i(y)\varphi_1, g_j(y)\varphi_2)$ is cyclic modulo $\gamma_{c+1}(A)$. By 6.1, the groups $\text{gp}(g_i(y)\varphi_1)$ and $\text{gp}(g_j(y)\varphi_2)$ are infinite cyclic modulo $\gamma_{c+1}(A)$, and so for nonzero $m_1, m_2 \in \mathbb{Z}$ we have

$$(g_i(y)\varphi_1)^{m_1} \equiv (g_j(y)\varphi_2)^{m_2} \not\equiv 1 \pmod{\gamma_{c+1}(A)}.$$

We have obtained a contradiction to Lemma 6.6, and the lemma is proved.

From this last assertion and 4.3 there follows immediately:

THEOREM 6.10. *Let $S = F_\infty(\mathfrak{N}_c)$. Then, for arbitrary $i \in \mathbb{Z}$, $w_i^4(S) \neq 1$ but $w_i^4(S) \cap \prod_{j \neq i} w_j^4(S) = 1$. Thus, for arbitrary odd $c \geq 9$, there is an infinite direct product of verbal subgroups in S .*

We set $W = \{w_i^4(t)\}_{i \in \mathbb{Z}}$. The following important fact follows from Theorem 6.10.

PROPOSITION 6.11. *There exists a relatively free 2-generator group $S_1 \in \mathfrak{A}^7$ whose verbal subgroup $W(S_1)$ is an infinite elementary abelian 2-group and has the following property: each endomorphism $\varphi \in \text{End } S_1$ acts like the identity map on $W(S_1)$, or else $W(S_1)\varphi = 1$.*

PROOF. Set $S = F_2(\mathfrak{N}_9)$. Carrying out for every word $w_i(t)$ arguments like those for v in the proof of Proposition 5.2, we construct verbal subgroups $H_i \subset w_i^4(S)$ such that $w_i^4(s_1, s_2) \notin H_i$, and for every $\varphi \in \text{End } S$ the following equation holds mod H_i :

$$w_i^4(s_1, s_2)\varphi \equiv \begin{cases} w_i^4(s_1, s_2), & \text{if } |\varphi| \neq 0, \\ 1, & \text{if } |\varphi| = 0. \end{cases}$$

We set $H = \prod_{i \in \mathbb{Z}} H_i$; then $S_1 = S/H$ is a relatively free group and $S_1 \in \mathfrak{A}_2^3 \mathfrak{N}_9 \subset \mathfrak{A}^7$. It follows from Theorem 6.10 that the images of the elements $\{w_i^4(s)\}$ under the natural epimorphism $S \rightarrow S_1$ constitute an additive basis of the elementary abelian 2-group $W(S_1)$. For arbitrary $\varphi \in \text{End } S_1$ we have $W(S_1)\varphi = 1$ when $|\varphi| = 0$, and $W(S_1)\varphi \equiv W(S_1)$ when $|\varphi| \neq 0$. This proves the proposition.

REMARK 6.12. In Theorem 6.10, the variety $\mathfrak{R}_c$ and the commutator $u_1(t)$ can be replaced by any variety $\mathfrak{R} \subseteq \mathfrak{R}_c$ and any word $v_1(t) \in \gamma_c(F_\infty)$ satisfying the following conditions: $R = F_\infty(\mathfrak{R})$ is torsion-free; $F_2(\mathfrak{R}) = F_2(\mathfrak{R}_c)$; and $v_1(R) \neq 1$ but $v_1(R) \cap u_2(R) \cdot u_3(R) = 1$.

§7. Some applications of Assertions 6.10 and 6.11

When we use the constructions from §6 in this section, we shall be assuming that the value of the parameter c is nine.

1°. We begin with answers to the problems of Kovács and Newman.

THEOREM 7.1. *There exists a center-by-metabelian variety $\mathfrak{Q}_1$ of exponent 8 that cannot be expressed as a union of two proper subvarieties and such that $\mathfrak{Q}_1\mathfrak{R}_9$ has continuum many decompositions as unions of two proper subvarieties.*

PROOF. We consider $L = F_\infty(\mathfrak{Q})$. By Zorn's lemma we can take a verbal subgroup V of L which is maximal with respect to having trivial intersection with L^4 . Set $\mathfrak{Q}_1 = \text{var}(L/V)$ and $L_1 = F_\infty(\mathfrak{Q}_1)$; then, by 3.8, any two nontrivial verbal subgroups of L_1 have nontrivial intersection. This is equivalent to the fact that $\mathfrak{Q}_1$ cannot be expressed as a union of two proper subvarieties. By Theorem 6.10, there is an infinite direct product of verbal subgroups $W(S_1) = \prod_{i \in \mathbb{Z}} w_i^4(S_1)$ in $S_1 = F_\infty(\mathfrak{Q}_1\mathfrak{R}_9)$. We take any subset $P \subseteq \mathbb{Z}$; then $W(S_1)$ can be expressed as the direct product of two subgroups $W(S_1) = \prod_{i \in P} w_i^4(S_1) \times \prod_{j \in \mathbb{Z} \setminus P} w_j^4(S_1)$. The corresponding decomposition of $\mathfrak{Q}_1\mathfrak{R}_c$ can be extended to a minimal one. It is clear that the minimal decompositions obtained in this way will be different for different P . This proves the theorem.

2°. From Theorem 6.10 there follows the independence of the system of identities $\{w_i^4(t) \equiv 1\}_{i \in \mathbb{Z}}$ in $\mathfrak{A}^7$. A stronger fact is true.

THEOREM. *The system of identities $\{w_i^4(t) \equiv 1\}_{i \in \mathbb{Z}}$ is 2-independent in $\mathfrak{A}^7$.*

PROOF. For arbitrary $i_1, \dots, i_k \in \mathbb{Z}$, we set $V = \{w_{i_1}^4(t), \dots, w_{i_k}^4(t)\}$, and suppose that $V(F_\infty) = \prod_1^n v_i(F_\infty)$ for some words $v_i(t), \dots, v_n(t) \in F_2$. We go over to the group S_1 of Proposition 6.11. The subgroup $V(S_1)$ is an elementary abelian 2-group of order 2^k . Let z_1 and z_2 be free generators of S_1 . Every subgroup of $V(S_1)$ is verbal in S_1 , so that $V(S_1) = \text{gp}(v_1(z), \dots, v_n(z))$. But then $|V(S_1)| \leq 2^n$, whence $n \geq k$. This proves the theorem.

3°. THEOREM 7.3. *Let $\mathfrak{M}$ be the subvariety defined in $\mathfrak{A}^7$ by the system of laws $\{w_i^4(t) \equiv 1\}_{i \in \mathbb{Z}}$. Then there exist continuum many different varieties lying in $\mathfrak{A}^7$ and covering $\mathfrak{M}$.*

PROOF. Set $G = F_\infty(\mathfrak{A}^7)$. To prove the theorem, it is enough for us to construct a continuum of maximal verbal subgroups of $W(G)$ (maximal in G , that is). For any subset $J \subset \mathbb{Z}$, we denote by $U(J)$ the verbal subgroup of G generated by the system of words

$$T(J): \{ w_j^4(t), w_q^4(t)w_p^{-4}(t) \mid j \in J: p \in \mathbb{Z} \setminus J \}.$$

We consider the group S of Proposition 6.11. Every subgroup of $W(S_1)$ is verbal in S_1 ; therefore for an arbitrary subset $J \subset \mathbb{Z}$ the verbal subgroup of S_1 generated by the system

of words $T(J)$ is of index 2 in $W(S_1)$ and does not contain the verbal subgroup $w_q^4(S_1)$ ($q \in \mathbb{Z} \setminus J$). Since $S_1 \in \mathfrak{A}^7$, it follows from this that $U(J) \not\supseteq w_q^4(G)$ ($q \in \mathbb{Z} \setminus J$). Let $r_1, r_2, \dots$ be free generators for G . Since $U(J) \not\supseteq w_q^4(r)$ ($q \in \mathbb{Z} \setminus J$), by Zorn's lemma we can choose for some fixed $q \in \mathbb{Z} \setminus J$ a verbal subgroup $V(J)$ of G such that $W(G) \supsetneq V(G) \supseteq U(J)$ and maximal with respect to not containing $w_q^4(r)$. For every proper subset $J \subset \mathbb{Z}$ we construct such a subgroup $V(J)$, and prove that the system of subgroups $\{V(J) \mid J \subset \mathbb{Z}\}$ is what we want.

We prove first that $V(J_1) \neq V(J_2)$ if $J_1 \neq J_2$. By construction $V(J) \subsetneq W(G)$; therefore it is enough for us to prove that $V(J_1) \cdot V(J_2) = W(G)$ if $J_1 \neq J_2$. Suppose, for example that $i \in J_1 \setminus J_2$ (which means that $i \in \mathbb{Z} \setminus J_2$). Then we get

$$V(J_1) \cdot V(J_2) \supset w_i^4(G) \cdot \prod_{j \in J_2} w_j^4(G) \cdot \prod_{g \in \mathbb{Z} \setminus J_2} (w_q^4 w_i^{-4})(G) = \prod_{k \in \mathbb{Z}} w_k^4(G) = W(G).$$

The maximality of the subgroups $V(J)$ in $W(G)$ follows since the inclusion $W(G) \supseteq V \supsetneq V(J)$, where V is verbal in G , implies that $V \ni w_p^4(r)$ for some $p \in \mathbb{Z} \setminus J$, and therefore that

$$V \supset w_p^4(G) \cdot V(J) = w_p^4(G) \cdot \prod_{j \in J} w_j^4(G) \cdot \prod_{q \in \mathbb{Z} \setminus J} (w_q^4 w_p^{-4})(G) = W(G).$$

This proves the theorem.

COROLLARY 7.4. *For any class $\mathcal{K}$ of groups of cardinality less than that of the continuum, there exist continuum many different varieties of solvable groups of length 7 with one and the same store of $\mathcal{K}$ -groups, and thus not generated by their $\mathcal{K}$ -groups.*

PROOF. Let $\mathfrak{M}$ be the variety in Theorem 7.3, and let $\mathfrak{A}^7 \supset \{\mathfrak{M}_\alpha\}$ be a system of continuum many varieties covering $\mathfrak{M}$. Since $\mathfrak{M}_{\alpha_1} \cap \mathfrak{M}_{\alpha_2} = \mathfrak{M}$ for $\alpha_1 \neq \alpha_2$, $(\mathfrak{M}_{\alpha_1} \setminus \mathfrak{M}) \cap (\mathfrak{M}_{\alpha_2} \setminus \mathfrak{M}) = \emptyset$ for $\alpha_1 \neq \alpha_2$. Thus we get a continuous system of nonintersecting classes $\{\mathfrak{M}_\alpha \setminus \mathfrak{M}\}$ of groups. From this system we choose continuum many classes having empty intersection with $\mathcal{K}$. Then for the corresponding system of varieties $\{\mathfrak{M}_\beta\}$ we have $\mathfrak{M}_\beta \cap \mathcal{K} \subset \mathfrak{M}$, and $\mathfrak{M}_{\beta_1} \cap \mathfrak{M}_{\beta_2} = \mathfrak{M}$ if $\beta_1 \neq \beta_2$. This concludes the proof.

COROLLARY 7.5. *For any class $\mathcal{K}$ of groups of cardinality less than that of the continuum, there is a 2-generator relatively free solvable group of length 7 that is not residually in $\mathcal{K}$.*

PROOF. This follows from 7.4 and the fact that the system of words $\{w_i^4(t)\}$ used in the proof of Theorem 7.5 depends on just two variables.

COROLLARY 7.6. *There exists a continuum of varieties of solvable groups of length 7 not generated by their recursively presented groups and thus not by groups with solvable word problem.*

§8. On the independence of systems of group laws

1°. We begin with a simple but important necessary condition for an independently based variety.

LEMMA 8.1. *Suppose the variety $\mathfrak{M}$ is independently based (in the variety $\mathfrak{B}$). Then for every variety $\mathfrak{M}_1 > \mathfrak{M}$ that is finitely based in $\mathfrak{B}$, there exists a subvariety $\mathfrak{M}_2 \subseteq \mathfrak{M}_1$ covering $\mathfrak{M}$.*

PROOF. Let $\{v_i(t) \equiv 1\}$ be an independent system of identities defining $\mathfrak{M}$ in $\mathfrak{B}$; let V be the verbal subgroup of $B = F_\infty(\mathfrak{B})$ corresponding to $\mathfrak{M}$, and $v(t) \equiv 1$ an identity defining the subvariety $\mathfrak{M}_1$ of $\mathfrak{B}$. By assumption, $V \supset v(B)$. If V is generated by a finite set of words, then by Zorn's lemma there exists a maximal verbal subgroup $V_2 \subseteq V$ (contained in B) such that $V_2 \supseteq v(B)$. Clearly, the subvariety $\mathfrak{M}_2$ of $\mathfrak{B}$ corresponding to V_2 will be what we want. If the system $\{v_i(t) \equiv 1\}$ is infinite, then for some n we have $v(B) \subseteq \prod_{i=1}^n v_i(B)$. Using Zorn's lemma, for V_2 we can choose a maximal verbal subgroup of B lying in V and containing $\prod_{i=n+1}^\infty v_i(B)$. This proves the lemma.

There is also a simple sufficient condition for a variety to be independently based: if the variety $\mathfrak{M}$ is determined in $\mathfrak{M}_1$ by an infinite independent system of identities $\{v_i(t) \equiv 1\}_{i \geq 1}$, then $\mathfrak{M}$ is independently based; that is, relative infinite independent basedness implies absolute (infinite) independent basedness. Indeed, if $\{r_i(t) \equiv 1\}$ is a basis for the laws of M_1 , then $\{v_i(t)r_i(t) \equiv 1\}$ is an independent basis for the identities of $\mathfrak{M}$ (the words v_i and r_i are written in disjoint sets of variables).

It is interesting that this condition is not necessary.

THEOREM 8.2. *There exist varieties $\mathfrak{M} \subset \mathfrak{M}_1 \subset \mathfrak{A}^7$ such that $\mathfrak{M}$ is independently based but does not have a covering variety in $\mathfrak{M}_1$; this means that it is not defined in $\mathfrak{M}_1$ by an independent system of identities.*

Before embarking on the proof of the theorem, we set

$$A = F_2(\mathfrak{A}_2 \mathfrak{N}_9) = \text{gp}(y_1, y_2), \quad R = A/\gamma_{10}(A) = F_2(\mathfrak{N}_9) = \text{gp}(x_1, x_2).$$

We choose for the $w_n(t)$ the words defined in §6.1° with $c = 9$, and we prove the following auxiliary assertion.

LEMMA 8.3. *For arbitrary endomorphisms $\varphi_1, \varphi_2 \in \text{End } A$, and any $i \in \mathbb{Z}$, the relation $w_i(y)\varphi_1 = w_i(y)\varphi_2$ implies that $\exp_2|\varphi_1| = \exp_2|\varphi_2|$.*

PROOF. By Lemmas 1.1–1.3, any two elements of A that are independent modulo A' are free $\mathfrak{A}_2 \mathfrak{N}_9$ -free generators for the subgroup A they generate. From this and Lemma 6.1 it follows that, for arbitrary $\varphi \in \text{End } A$, $w_i(y)\varphi = 1$ if and only if $|\varphi| = 0$. Therefore we may assume that $|\varphi_1||\varphi_2| \neq 0$. Lemma 6.8 allows us to conclude that $\text{gp}(g_i(y)\varphi_1, g_i(y)\varphi_2)$ is infinite cyclic modulo $\gamma_{10}(A)$. Suppose that $v(y)$ is not a proper power modulo $\gamma_{10}(A)$ and generates this group; that is,

$$v(y)^{m_1} \equiv g_i(y)\varphi_1 \not\equiv 1, \quad v(y)^{m_2} \equiv g_i(y)\varphi_2 \not\equiv 1 \pmod{\gamma_{10}(A)}.$$

Then using the Magnus-Shmel'kin embedding (§6.3°) and going over to the wreath product, we get the following equation in the module M (see the proof of Lemma 6.7):

$$u_0(y)\varphi_1\mu(v(x)^{m_1} - 1) = u_0(y)\varphi_2\mu(v(x)^{m_2} - 1). \quad (8.1)$$

For arbitrary $m \in \mathbb{Z}$ the obvious equation

$$v(x)^m - 1 = \alpha \cdot (v(x) - 1)^{2^{\exp_2 m}}$$

holds, where $\alpha \in \mathbb{Z}_2(R) \setminus \Delta_R$. Therefore (8.1) acquires the form

$$u_0(y)\varphi_1\mu \cdot \alpha_1 \cdot (v(x) - 1)^{2^{\exp_2 m_1}} = u_0(y)\varphi_2\mu \cdot \alpha_2 (v(x) - 1)^{2^{\exp_2 m_2}}, \quad (8.2)$$

where $\alpha_1, \alpha_2 \in \mathbb{Z}_2(R) \setminus \Delta_R$.⁽⁴⁾

⁽⁴⁾ Expression (8.2) shows that the parameter $\exp_2|\varphi|$ reflects the "depth of the bedding" of $w_i(y)\varphi$ in A ; that is, it grows with the growth in the numeral of the term of the lower central series of A containing this element.

We assume that $\exp_2 m_1 \neq \exp_2 m_2$ (for definiteness suppose that $\exp_2 m_1 > \exp_2 m_2$); then on cancelling (8.2) by the factor $(v(x) - 1)^{2\exp_2 m_2}$ we get that $u_0(y)\varphi_2\mu \cdot \alpha_2 \in M \cdot \Delta_R$. By Lemma 6.7, the contradiction thus obtained proves that $\exp_2 m_1 = \exp_2 m_2$. But by Lemmas 6.5 and 6.4, $m_1^{1/9}(\varphi_1) = m_1^{1/9}(\varphi_2)$, whence it follows that $\exp_2(m_1^2|\varphi_1|^9) = \exp_2(m_1^2|\varphi_2|^9)$, and this means that $\exp_2|\varphi_1| = \exp_2|\varphi_2|$. This proves the lemma.

PROOF OF THEOREM 8.2. We set $S = F_2(\mathfrak{N}_9)$. For $\mathfrak{M}$ we take the subvariety of $\mathfrak{N}_9$ defined by the system of identities $\{w_i^4(t) \equiv 1\}_{i \geq 0}$. By 6.10, $\mathfrak{M}$ is independently based. We set $\mathfrak{M} = W(S)$ (where $W = \{w_i^4(t)\}_{i \geq 0}$) and $M_i = w_i^4(S)$. To prove the theorem, it is enough if we find a verbal subgroup V of S lying in M such that for every verbal subgroup U with $M \supsetneq U \supseteq V$ there exists an intermediate verbal subgroup U_1 such that $M \supsetneq U_1 \supsetneq U \supseteq V$. In fact, since $\mathfrak{M}$ is defined in $\mathfrak{N}_9$ by words in two variables, on transition from the subgroups $W(S)$ and V to the verbal subgroups $W(F_\infty(\mathfrak{N}_9))$ and $V(F_\infty(\mathfrak{N}_9))$, these last preserve the same property. And this means precisely that the subvariety $\mathfrak{M}_1$ of $\mathfrak{N}_9$ corresponding to V does not contain a variety covering $\mathfrak{M}$.

As in §5, we consider the module interpretation of the subgroups M and M_i . By 6.10, we have $M = \bigoplus_0^\infty M_i$ for the corresponding modules. It is enough for us to find a submodule $V \subsetneq M$ not contained in any proper submodule of M .

Each M_i is a cyclic right module over $\mathbf{Z}_2(\text{End } S)$ with generator $w_i^4(s)$. As in the proof of Proposition 5.2, we prove that the ring $\mathbf{Z}_2(\text{End } S)$ may be replaced by $\mathbf{P} = \mathbf{Z}_2(Q)$, where $Q = \{\varphi \in \text{End } S \mid |\varphi| \neq 0\}$, and that in this transition every P -module is a $\mathbf{Z}_2(\text{End } S)$ -submodule.

We take the two-sided ideal T of P generated by all elements of the form $\{\varphi_1 - \varphi_2|\varphi_1, \varphi_2 \in Q; \exp_2|\varphi_1| = \exp_2|\varphi_2|\}$, and set $N_i = M_i \cdot T$. By Lemmas 4.1, 5.1 and 8.3, $\text{Ann}_P(w_i^4(s)) \subset T$. Therefore the module homomorphisms $P \rightarrow P/\text{Ann}_P(w_i^4(s)) \cong M_i$ can be extended to homomorphisms $P \rightarrow P/T \cong M_i/N_i$. The image of the semigroup Q in the quotient ring P/T is obviously an infinite cyclic semigroup with identity, with generator a coset of the form $\varphi + T$, $\exp_2|\varphi| = 1$.

We fix an arbitrary endomorphism $\theta \in \text{End } S$ such that $\exp_2|\theta| = 1$. Then there exists a natural isomorphism from P/T onto the ring $\mathbf{Z}_2[\theta]$ of polynomials, established as follows: to every element of P of the form $\varphi_1 + \dots + \varphi_n$ ($\varphi_i \in Q$) there corresponds the polynomial $\sum_1^n \theta^{\exp_2|\varphi_i|}$. It follows from this that all the quotient modules M_i/N_i are isomorphic with a free cyclic module over $\mathbf{Z}_2[\theta]$ (in this transition the correspondence between verbal subgroups and submodules continues to survive).

Suppose now that N is the submodule of M generated by all elements of the form $\{w_i^4(s)\theta - w_{i-1}^4(s) \mid i \geq 1\}$. We define the submodule V as $V = N + M_0 + (\bigoplus_1^\infty N_i)$, and show that it is the one we want. We consider the quotient module M/V . The endomorphism $M \rightarrow M/V$ can be factored through the epimorphism $M \rightarrow M/\bigoplus_1^\infty N_i \cong P/T \oplus \dots \oplus P/T \oplus \dots$. Denote by r_i the image of $w_i^4(s)$ under the homomorphism $M \rightarrow M/\bigoplus_1^\infty N_i$. Then, from what has been said above,

$$M/V \cong \left\langle \bigoplus_{i=0}^\infty r_i \cdot \mathbf{Z}_2[\theta] \mid r_0 = 0, r_j\theta = r_{j-1}, j \geq 1 \right\rangle.$$

It follows in turn from this that every proper submodule of M/V is the cyclic submodule generated by some r_i . In fact, using the "shift" multiplication by θ , it is easy to prove that, together with every element of the form $r_{i_1} + \dots + r_{i_k}$ ($i_1 < \dots < i_k$), a submodule contains each of the elements $r_1, \dots, r_{i_k}$. Since the whole quotient module M/V is

obviously not finitely generated, it does not contain a proper maximal submodule. This concludes the proof of the theorem.

REMARK 8.4. The verbal subgroup V of S corresponding to the variety $\mathfrak{M}_1$ of Theorem 8.2 is defined in S by the following system of relations:

$$\{w_0^4(t) \equiv 1, w_i^4(s)\varphi_1 = w_i^4(s)\varphi_2, w_i^4(s)\varphi = w_{i-1}^4(s)\varphi \mid i \geq 1, \\ \times \varphi, \varphi_1, \varphi_2 \in \text{End } S; \exp_2|\varphi_1| = \exp_2|\varphi_2|, \exp_2|\varphi| = 1\}.$$

It is interesting to note that the verbal subgroup $W(F_2(\mathfrak{M}_1))$ is infinite, but at the same time every verbal subgroup of $F_2(\mathfrak{M}_1)$ contained properly in it is finite and indeed is the fully invariant closure of a single element. This follows from the fact that every proper submodule K of M/V is a cyclic module over $\mathbb{Z}_2[\theta]$ with generator r_i , say, and if $r_j\theta = r_{j-1}$ ($j \geq 1$) in this case, then $|K| = 2^i$.

2°. In this subsection we prove the main assertion of the section.

THEOREM 8.5. *There exists a variety of solvable groups of length 7 that is not independently based.*

The identities giving the variety in Theorem 8.5 will be written out explicitly. As follows from 8.1, for the proof it is enough to indicate a finitely based variety $\mathfrak{M}_1 \subset \mathfrak{M}^7$ containing a subvariety $\mathfrak{M}$ not having a covering subvariety in $\mathfrak{M}_1$. The following assertion shows the approach to the solution of this question.

LEMMA 8.6 (on the shift endomorphism). *Let $\mathfrak{M}_1$ be a finitely based variety, suppose that $G = F_2(\mathfrak{M}_1)$ is freely generated by elements z_1, z_2 , and let $V = \{v_i(t_1, t_2)\}_{i \geq 1}$ and $\varphi \in \text{End } G$. If $V(G) \neq 1$ and G satisfies the following conditions (where for convenience we set $1 = v_0(z)$):*

- a) $v_i(z)\varphi = v_{i-1}(z)$,
- b) $v_i(G)\varphi \subseteq v_{i-1}(G)$,

then the subvariety defined in $\mathfrak{M}_1$ by the system of identities $\{v_i(t) \equiv 1\}_{i \geq 1}$ does not have a covering variety in $\mathfrak{M}_1$ and thus is not independently based.

PROOF. Since V is a set of words depending on just two variables, it is enough if we prove that $V(G)$ does not contain proper subgroups that are maximal verbal in G . Assume the converse, and denote such a subgroup by V_1 ; then $V(G) \equiv v_n(G) \pmod{V_1}$ for some $n \geq 1$. It follows from property a) that $V(G)\varphi^n \not\subseteq V_1$ for each $n \geq 1$, and from property b) that $v_n(G)\varphi^n = 1$. Thus, applying φ^n to both sides of the congruence, we get the contradiction

$$V(G)\varphi^n \equiv v_n(G)\varphi^n \equiv 1 \pmod{V_1}.$$

This proves the lemma.

We model the phenomenon occurring in Lemma 8.6 in a certain factor-group of $S = F_2(\mathfrak{M}_{11}) = \text{gp}(s_1, s_2)$. First we introduce some notation. We set

$$A = F_2(\mathfrak{M}_2\mathfrak{M}_{11}) = \text{gp}(y_1, y_2), \quad R = F_2(\mathfrak{M}_{11}) = \text{gp}(x_1, x_2)$$

and consider any homomorphism $\varphi \in \text{Hom}(F_4, R)$ ($\varphi \in \text{Hom}(F_4, A)$). The matrix of (φ) takes the form

$$(\varphi) = \begin{pmatrix} a & b \\ p & q \\ k & l \\ m & n \end{pmatrix}.$$

We set

$$A_\varphi = \begin{pmatrix} a & b \\ p & q \end{pmatrix}, \quad B_\varphi = \begin{pmatrix} k & l \\ m & n \end{pmatrix}.$$

We set $v_0(t) = [[t_2, t_1, t_2], [t_2, t_1, t_1], [t_2, t_1, t_1]]$ and define these new words in F_4 :

$$v(t) = [v_0(t), [t_4, t_3]], \quad g(t) = v(t)u_2(t)u_3(t), \quad w(t) = [u_0(t), g(t)],$$

where the words $u_0(t)$, $u_2(t)$ and $u_3(t)$ of §6.1° correspond to the value $c = 11$ of c . We recall the following notation of §6:

$$u_1(t) = [v_0(t), [t_2, t_1]], \quad g_n(t) = u_1^n(t)u_2(t)u_3(t), \quad w_n(t) = [u_0(t), g_n(t)].$$

We consider how the values of $v(t)$ appear in R :

$$v(t)\varphi = [v_0(t)\varphi, [t_4, t_3]\varphi] = [v_0(t)\varphi, [x_2, x_1]^{B_\varphi}] = [v_0(t)\varphi, [x_2, x_1]^{B_\varphi}].$$

In particular, $v(x_1, x_2, x_1, x_2^n) = u_1^n(x)$, which means that $g(x_1, x_2, x_1, x_2^n) = g_n(x)$. Since $\gamma_{12}(A)$ is abelian and $u_0(A) \subset \gamma_{12}(A)$, we have

$$w(y_1, y_2, y_1, y_2^n) = w_n(y),$$

which together with 4.1 gives

$$w^4(s_1, s_2, s_1, s_2^n) = w_n^4(s). \quad (8.3)$$

The last expression proves that $w^4(s) \supset \prod_{i \in \mathbb{Z}} w_i^4(s)$.

3°. In order to create the situation described in Lemma 8.6, we impose four identities on S . For the first one we take the identity (T1) defining $\mathfrak{S}\mathfrak{R}_{11}$ (see Lemma 4.6).⁽⁵⁾ Since it defines S , it does not impose any further relations on it.

Before defining the second identity, we note that the action of $\varphi \in \text{End } S$ on the right-hand side of $w_n^4(s) = [u_0(s), g_n(s)]^4$ is uniquely determined by the matrix (φ) . In other words, the following equation holds in S for arbitrary elements $r_1 = s_1^{a_1}s_2^{b_1}$, $r_2 = s_1^{a_2}s_2^{b_2}$ and $d_1, d_2 \in S'$:

$$[u_0(r_1d_1, r_2d_2), g_n(r_1d_1, r_2d_2)]^4 = [u_0(r_1d_1, r_2d_2), g_n(r_1, r_2)]^4. \quad (8.4)$$

This follows immediately from the distributivity in every place of commutators of weight 11 in R , the fact that $\gamma_{12}(A)$ is abelian, and Lemma 4.1.

We want imposition of the second identity to be such that modulo it the action of any endomorphism φ on the words $w_n^4(s)$ is uniquely determined by the matrix (φ) , or, in other words, such that the action of φ on the $w_n^4(s)$ is defined by the "linear part" of φ . As is easy to see, this is equivalent to the requirement that, modulo the imposed identity, the marginal subgroup $(w_n^4)^*(S)$ contains the derived group of S . Clearly, in view of (8.4) this is so provided that all equations of the following form hold in S :

$$[u_0(r_1d_1, r_2d_2), g_n(r_1, r_2)]^4 = [u_0(r_1, r_2), g_n(r_1, r_2)]^4. \quad (8.4')$$

Thus for (T2) we may take the identity

$$\begin{aligned} (T2): \quad & [u_0(t_1 \cdot [t_5, t_6], t_2 \cdot [t_7, t_8]), g(t_1, t_2, t_3, t_4)]^4 \\ & \equiv [u_0(t_1, t_2), g(t_1, t_2, t_3, t_4)]^4 = w^4(t), \end{aligned}$$

⁽⁵⁾ This identity has the form

$$(((r_1r_2)^2(r_3r_4)^2)^2((r_5r_6)^2(r_7r_8)^2)^2) \cdot [((r_1r_2)^2(r_3r_4)^2)^2, r_9] = 1,$$

where the r_i are copies of $[t_1, \dots, t_{12}]$ written in disjoint sets of variables.

imposition of which turns the derived group of S into the marginal subgroup $(w^4)^*(S)$, and in view of 8.3, it turns it into the marginal subgroup $(w_n^4)^*(S)$ for all $n \in \mathbf{Z}$.

The following identity guarantees for us the analogue of the equation $v_0(z) = v_1(z)\varphi = 1$ as in Lemma 8.6:

$$(T3): w_1^4(t) \equiv 1.$$

The last identity guarantees shift of the chain of words:

$$(T4): w^4(t_1, t_2, t_3, t_4) \equiv w^4(t_1^2, t_2^2, t_3^2, t_4^2).$$

Let G be the group obtained from S by adding $(T2)$ – $(T4)$, and let z_1 and z_2 be free generators for G . We set $V = \{w_{2^n}^4(t)\}_{n \geq 1}$, and denote by $(T5)$ the corresponding system of identities $\{w_{2^n}^4(t) \equiv 1\}_{n \geq 1}$. For φ we take the endomorphism defined by $\varphi: z_i \rightarrow z_i^2$ ($i = 1, 2$).

LEMMA 8.7. *For arbitrary $n \geq 2$, the following conditions hold in G :*

- a) $w_{2^n}^4(z)\varphi = w_{2^{n-1}}^4(z)$, $w_2^4(z)\varphi = 1$;
- b) $w_{2^n}^4(G)\varphi \subset w_{2^{n-1}}^4(G)$, $w_2^4(G)\varphi = 1$.

PROOF. First of all we note that modulo $(T2)$, condition a) implies b). For, b) holds if for arbitrary $n \geq 2$ and $\psi \in \text{End } G$ we have $w_{2^n}^4(z)\psi\varphi \in w_{2^{n-1}}^4(G)$ and $w_2^4(z)\psi\varphi = 1$. But because of the choice of $(T2)$, the action of every endomorphism on $w_{2^n}^4(z)$ is uniquely determined by its matrix. Since (φ) is a scalar matrix, $(\psi\varphi) = (\psi)(\varphi) = (\varphi)(\psi) = (\varphi\psi)$, which means that $w_{2^n}^4(z)\psi\varphi = w_{2^n}^4(z)\varphi\psi \stackrel{a)}{=} w_{2^{n-1}}^4(z)\psi$ and $w_2^4(z)\psi\varphi = w_2^4(z)\varphi\psi \stackrel{a)}{=} 1$. Thus it is enough if we prove that condition a) holds. But

$$\begin{aligned} w_{2^n}^4(z)\varphi &\stackrel{(8.3)}{=} w^4(z_1, z_2, z_1, z_2^{2^n})\varphi = w^4(z_1^2, z_2^2, z_1^2, z_2^{2^{n+1}}) \\ &\stackrel{(T4)}{=} w^4(z_1, z_2, z_1, z_2^{2^{n-1}}) \stackrel{(8.3)}{=} w_{2^{n-1}}^4(z). \end{aligned}$$

In particular, $w_2^4(z)\varphi = w_1^4(z) \stackrel{(T3)}{=} 1$. This proves the lemma.

To prove the theorem we need another auxiliary assertion.

LEMMA 8.8. *Suppose that $\varphi_1, \varphi_2 \in \text{Hom}(F_4, A)$. If $w(t)\varphi_1 = w(t)\varphi_2 \neq 1$, then*

$$\exp_2|A_{\varphi_1}| = \exp_2|A_{\varphi_2}|, \quad \exp_2|B_{\varphi_1}| = \exp_2|B_{\varphi_2}|.$$

PROOF. We may assume that $|A_{\varphi_1}| \cdot |A_{\varphi_2}| \neq 0$. In fact, by 6.1, we have $g(t)\varphi \in \gamma_{12}(A)$ for every homomorphism $\varphi \in \text{Hom}(F_4, A)$ such that $|A_\varphi| = 0$, which means that $w(t)\varphi = 1$.

The homomorphisms φ_i may be represented in the form

$$\varphi_i: F_4 \xrightarrow{\psi} F_4(\mathfrak{A}_2 \mathfrak{R}_{11}) \xrightarrow{\varphi'_i} A,$$

where the $t_j\psi$ are free generators of $F_4(\mathfrak{A}_2 \mathfrak{R}_{11})$ ($j = 1, \dots, 4$), and $(\varphi_i) = (\varphi'_i)$. Therefore we may use Lemma 6.8, in view of which there exist m_1 and m_2 such that

$$(g(t)\varphi_1)^{m_1} \equiv (g(t)\varphi_2)^{m_2} \neq 1 \pmod{\gamma_{12}(A)}. \quad (8.5)$$

Since $R = A/\gamma_{12}(A)$ is torsion-free, we may assume that $(m_1, m_2) = 1$. Just as in the proof of Lemma 8.3, from (8.5) and the equation $w(t)\varphi_1 = w(t)\varphi_2$ we get $\exp_2 m_1 = \exp_2 m_2$. Further, since in Lemma 6.5 we actually proved that $v(R) \cap u_2(R)u_3(R) = 1$,

from (8.5) we get

$$((u_2(t)u_3(t))\varphi_1)^{m_1} \equiv ((u_2(t)u_3(t))\varphi_2)^{m_2} \pmod{\gamma_{12}(A)}, \quad (8.6)$$

$$(v(t)\varphi_1)^{m_1} \equiv (v(t)\varphi_2)^{m_2} \pmod{\gamma_{12}(A)}. \quad (8.7)$$

Lemma 6.4 and relation (8.6) imply that $m_1^{1/11}A_{\varphi_1} = m_2^{1/11}A_{\varphi_2}$, which together with the condition $(m_1, m_2) = 1$ allows us to assume in particular that $n_1 = m_1^{1/11}$ and $n_2 = m_2^{1/11}$ are integers. From this we get that $\exp_2(m_1^{2/11}|A_{\varphi_1}|) = \exp_2(m_2^{2/11}|A_{\varphi_2}|)$, which means that $\exp_2|A_{\varphi_1}| = \exp_2|A_{\varphi_2}|$.

Now set $\delta_2 = |B_{\varphi_2}|$; then the obvious equation $[t_4, t_3]\varphi_i \equiv [y_2, y_1]^{\delta_i} \pmod{A'}$ implies the equation

$$v(t)\varphi_i \equiv [v_0(t)\varphi_i, [y_2, y_1]^{\delta_i}] \equiv [v_0(t)\varphi_i, [y_2, y_1]]^{\delta_i} \pmod{\gamma_{12}(A)}.$$

We define the integer matrix $D = n_2^{-1}A_{\varphi_1} = n_1^{-1}A_{\varphi_2}$ and choose any homomorphism $\varphi \in \text{Hom}(F_4A)$ such that $A_\varphi = D$. Since $v_0(t) = v_0(t_1, t_2)$, we get

$$\begin{aligned} (v(t)\varphi_1)^{m_1} &\equiv [v_0(t)\varphi_1, [y_2, y_1]]^{m_1\delta_1} \equiv [(v_0(t)\varphi)^{n_2^2}, [y_2, y_1]]^{m_1\delta_1} \\ &\equiv [v_0(t)\varphi, [y_2, y_1]]^{m_1\delta_1 n_2^2} \pmod{\gamma_{12}(A)}. \end{aligned}$$

Analogously,

$$(v(t)\varphi_2)^{m_2} \equiv [v_0(t)\varphi, [y_2, y_1]]^{m_2\delta_2 n_1^2} \pmod{\gamma_{12}(A)}.$$

The last two equations, together with (8.7), tell us that $m_1\delta_1 m_2^{9/11} = m_2\delta_2 m_1^{9/11}$, whence follows the required relation $\exp_2 \delta_1 = \exp_2 \delta_2$. This proves the lemma.

PROOF OF THEOREM 8.5. By (8.6) and (8.7), it is enough if we prove that (T1)–(T4) do not imply (T5). We shall show that even the identity $w_2^4(t) \equiv 1$ does not follow from (T1)–(T4).

As a preliminary we note that, for every homomorphism $\varphi \in \text{Hom}(F_4, A)$, $w(t)\varphi = 1$ if and only if $|A_\varphi| = 0$. For it was proved in the proof of Lemma 8.8 that $w(t)\varphi = 1$ if $|A_\varphi| = 0$. Suppose that $|A_\varphi| \neq 0$; since the words $u_0(t)$, $u_2(t)$ and $u_3(t)$ lie in F_2 and $v(R) \cap u_2(R)u_3(R) = 1$, from 6.1 it follows that $g(t)\varphi \notin \gamma_{12}(A)$. Using the Magnus-Shmel'kin embedding and Lemma 6.7, we get that $w(t)\varphi \neq 1$.

Suppose now that $\varphi \in \text{Hom}(F_4, S)$. Then the preceding remark and Lemma 4.1 prove that $|A_\varphi| = 0$ if and only if $w^4(t)\varphi = 1$.

We consider the verbal subgroup $M = w^4(S)$; it is generated by all elements of the form $w^4(t)\varphi$ ($\varphi \in \text{Hom}(F_4, S)$), and by what was said above we may assume that $|A_\varphi| \neq 0$. With every homomorphism $\varphi \in \text{Hom}(F_4, S)$ such that $|A_\varphi| \neq 0$ we associate the pair of numbers

$$[\varphi] = (\exp_2|A_\varphi|, \exp_2|B_\varphi|).$$

The first component of a pair is finite; the second can be infinite. For arbitrary $\theta \in \text{End } S$ ($|\theta| \neq 0$) we have the obvious equation

$$[\varphi\theta] = (\exp_2|A_\varphi| + \exp_2|\theta|, \exp_2|B_\varphi| + \exp_2|\theta|). \quad (8.8)$$

By 4.1 and 8.8, the inequality $[\varphi_1] \neq [\varphi_2]$ implies that $w^4(t)\varphi_1 \neq w^4(t)\varphi_2$.

Let N be the subgroup of S generated by the set

$$\left\{ w^4(t)\varphi_1(w^4(t)\varphi_2)^{-1} \mid \varphi_1, \varphi_2 \in \text{Hom}(F_4, S); [\varphi_1] \neq [\varphi_2] \right\}.$$

By (8.8), N is fully invariant in S . We shall show that the verbal subgroup $\bar{M} = M/N$ of $\bar{S} = S/N$ can be considered as a free module over the polynomial ring $\mathbf{Z}_2[\theta]$ (where for θ we may take any endomorphism $\theta \in \text{End } \bar{S}$ such that $\exp_2|\theta| = 1$) with basis $B = \{(0, 0), (0, \infty), (i, 0), (0, i) | i = 1, 2, \dots\}$ consisting of the pairs of numbers indicated, and action $(m, n)\theta = (m + 1, n + 1)$ (by definition $\infty + 1 = \infty$); the submodules of this module are in one-to-one correspondence with the verbal subgroups of $\bar{S}$ lying in $\bar{M}$.

In fact, as was noted above, the elementary abelian 2-group M is generated by all elements of the form $w^4(t)\varphi$, where $\varphi \in \text{Hom}(F_4, S)$ and $|A_\varphi| \neq 0$. In general these elements are not independent generators of M . It follows from 4.2 that all relations among them are consequences of relations of the form $w^4(t)\varphi_1 = w^4(t)\varphi_2$, where $w(t)\varphi_1 \equiv w(t)\varphi_2 \pmod{x^2(\gamma_{12}(S))}$. It then follows from Lemma 8.8 that a set of independent generators of M modulo N is the set of elements $\{w^4(t)\varphi_i\}_{i \in I}$ such that for every pair (m, n) of the set of pairs $D = \{(m, n) | m = 0, 1, \dots; n = \infty, 0, 1, \dots\}$ there is one and only one $i \in I$ such that $[\varphi_i] = (m, n)$. In other words, modulo N the subgroup M can be viewed as a vector space over $\mathbf{Z}_2$ with D as basis. (For convenience we identify pairs of the form (∞, n) , setting them equal to zero.) By (8.8), the action of the endomorphism $\theta \in \text{End } S$ on M has the following form in this version:

$$(m, n)\theta = (m + \exp_2|\theta|, n + \exp_2|\theta|).$$

This means that the action of the semigroup $\text{End } \bar{S}$ on $\bar{M}$ degenerates to that of an infinite cyclic semigroup with zero and identity, with generator any endomorphism θ such that $\exp_2|\theta| = 1$, the role of the zero being played by endomorphisms whose matrices have zero determinants. Thus when the verbal subgroup $\bar{M}$ of $\bar{S}$ is viewed as a module over the semigroup ring $\mathbf{Z}_2(\text{End } \bar{S})$ (see §5), we get a free module over $\mathbf{Z}_2[\theta]$ with basis B . It is clear that this transition preserves the one-to-one correspondence between submodules and verbal subgroups of $\bar{S}$ lying in $\bar{M}$.

We note that $\bar{S}$ satisfies (T2). For, the arguments used in the construction of (T2) show that (T2) holds provided that the image of $w^4(t)$ under the action of an arbitrary homomorphism $\varphi \in \text{Hom}(F_4, \bar{S})$ is uniquely determined by the matrix (φ) (see (8.3)–(8.4')). As is easy to see, the last holds because the equation $(\varphi_1) = (\varphi_2)$, where $\varphi_i \in \text{Hom}(F_4, S)$, implies that $[\varphi_1] = [\varphi_2]$, which means that $w^4(t)\varphi_1 \equiv w^4(t)\varphi_2 \pmod{N}$.

We now impose the identities (T3) and (T4) on $\bar{S}$. By (8.3), the elements $w_2^4(S)$ ($n \geq 0$) in $\bar{M}$ corresponds to the generators $(0, n)$, so that clearly the first identity is equivalent to the relation $(0, 0) = 0$ in the module M . (T4) is equivalent to the system of relations

$$Q = \{ (m, n) = (m + 2, n + 3) | m = 0, 1, \dots; n = \infty, 0, 1, \dots \}$$

in $\bar{M}$. For, every value of $w^4(t)$ in $\bar{S}$ is an element of the form $w^4(t)\varphi$, where $\varphi \in \text{Hom}(F_4, \bar{S})$; in our notation, this is the element $(m, n) = [\varphi]$ of the module $\bar{M}$, while at the same time

$$w^4(t_1^2, t_2^2, t_1^2, t_2^4)\varphi = (\exp_2|A_\varphi| + 2, \exp_2|B_\varphi| + 3) = (m + 2, n + 3).$$

We have reduced the proof that the identity $w_2^4(t) \equiv 1$ is independent of (T1)–(T4) to checking that the relation $(0, 1) = 0$ in $\bar{M}$ does not follow from the system of relations $Q \cup \{(0, 0) = 0\}$.

Assume the contrary, that is, that there is a relation of the form

$$(0, 1) = \sum_j (m_j, m_j) + \sum_i ((m_i + 2, n_i + 3) - (m_i, n_i)) \quad (8.9)$$

in $\overline{M}$. Since all nonnegative pairs (m, n) form a basis of $\overline{M}$ as a vector space over $\mathbb{Z}_2$, the right-hand side of (8.9) must contain the pair $(0, 1)$, and clearly it must occur in the second sum. But then for some nonempty set of nonzero natural numbers $\{k_\alpha\}$, summands of the form $(2k_\alpha, 3k_\alpha + 1)$ must occur in the right-hand side of (8.9). It is clear that the summand of this sort with maximal coordinates cannot be cancelled with anything, and thus (8.9) does not hold. This contradiction concludes the proof of the theorem.

REMARK 8.9. We have established in the proof of Theorem 8.5 that the identities (T1)–(T5) define a variety $\mathfrak{M}$ of solvable groups of length 7 that is not independently based. It is not hard to show that $\mathfrak{M}$ can be defined by identities in 89 variables. Further, it follows from 6.10 that the system (T3)–(T4) is independent (modulo (T1), even). From the proof of Theorem 8.5 and a few additional arguments one can deduce that the system (T1)–(T3), (T5) is also independent. (Thus for instance the independence of (T1) from (T2)–(T5) follows from Razmyslov's result [25] on the nonsolvability of $\mathfrak{B}_4$.) Therefore $\mathfrak{M}$ is the intersection of an independently based variety and a variety defined by a single identity in four variables.

COROLLARY 8.10. *The intersection of a finitely based variety and an independently based variety need not be independently based; so the intersection of two independently based varieties may be not independently based.*

We denote by $v_i(t) \equiv 1$ an identity defining the variety defined by the identities (T1)–(T4) and $w_2^4(t) \equiv 1$. It is easy to show that the words $v_i(t)$ can be written in terms of 89 variables, so that from the proof of Theorem 8.5 we immediately get

COROLLARY 8.11. *The set of words $\{v_i(t)\}_{i \geq 1} \subset F_{89}$ has the following property: the chain of verbal subgroups $v_1(F_{89}) \subsetneq v_2(F_{89}) \subsetneq \cdots \subsetneq v_n(F_{89}) \subsetneq \cdots$ is strictly increasing.*

Some additional arguments (use of $\mathfrak{R}_9$ instead of $\mathfrak{R}_{11}$) allow us to reduce the number of variables in Corollary 8.1 to 73.

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